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LaED: a novel lightweight, edge-aware and explainable deep learning model for privacy-preserving facial attendance tracking in resource-constrained educational environments

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Facial recognition is increasingly adopted for automated classroom attendance; however, real-world deployment in schools remains constrained by privacy risks, ethical obligations, demographic bias, spoofing threats, and limited computational resources. Recent incidents involving Microsoft Teams in New South Wales in 2025 and Chelmer Valley High School in the United Kingdom show how poorly governed systems violate student rights and regulatory compliance. Despite growing adoption, many existing attendance systems focus narrowly on recognition accuracy or efficiency, while overlooking spoof resistance, open-set identity handling, fairness mitigation, auditability, and privacy protection. This paper presents LaED, a lightweight, edge-aware, and explainable deep learning framework for privacy-preserving classroom attendance in resource-constrained educational environments. The framework combines multimodal spoof detection, open-set facial recognition, and fairness-aware representation learning within a unified edge-based design. Spoofing attacks, including replay and deepfake attempts, are mitigated through the fusion of physiological and temporal facial cues, while unknown identities are explicitly rejected to reduce proxy attendance. To support responsible deployment, LaED incorporates federated learning with differential privacy, ensuring that biometric data remain local to schools while enabling accountable model updates. Experimental evaluation on CASIA-FASD, CelebA-Spoof, DFDC, FairFace, and a consent-driven classroom dataset shows that LaED achieves over 97.8% recognition accuracy, APCER and BPCER values below 2%, demographic fairness gaps under 2%, and inference latency below 150 milliseconds on edge hardware. Additional tests confirm reliable operation under realistic classroom conditions. These results demonstrate that regulation-aligned and trustworthy facial attendance is feasible on low-cost devices, offering a practical pathway for responsible biometric AI in education.

Keywords Privacy-preserving facial recognition, Classroom attendance tracking, Lightweight deep learning, Federated learning with differential privacy, Fairness and explainability in AI, Edge intelligence, Green AI

Abbreviations

AdaFace	Adaptive Margin Face Recognition Model
AI	Artificial Intelligence
APCER	Attack Presentation Classification Error Rate
API	Application Programming Interface
ArcFace	Additive Angular Margin Loss for Face Recognition
AUC	Area Under the Curve

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BBC	British Broadcasting Corporation
BPCER	Bona Fide Presentation Classification Error Rate
CASIA-FASD	CASIA Face Anti-Spoofing Database
CI	Confidence Interval
CNN	Convolutional Neural Network
DFDC	DeepFake Detection Challenge
DNN	Deep Neural Network
EER	Equal Error Rate
ElasticFace	Elastic Margin Face Recognition Model
FAR	False Acceptance Rate
FGSM	Fast Gradient Sign Method
FNMR	False Non-Match Rate
FR	Face Recognition
GAN	Generative Adversarial Network
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
IoT	Internet of Things
LaED	Lightweight, Edge-aware and Explainable Deep Learning framework (proposed)
LFW	Labeled Faces in the Wild
MagFace	Magnitude-Aware Face Recognition Model
ML	Machine Learning
NIR	Near-Infrared
NSW	New South Wales
OpenFace	OpenFace Recognition Model
PCA	Principal Component Analysis
PGD	Projected Gradient Descent
RAM	Random Access Memory
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
SUS	System Usability Scale
SVM	Support Vector Machine
StyleGAN3	Alias-Free Generative Adversarial Network for Image Synthesis
TPR	True Positive Rate
TPU	Tensor Processing Unit
ViT-FaceNet	Vision Transformer variant of FaceNet

Facial recognition technologies have rapidly gained traction for automating administrative tasks across diverse domains, including law enforcement, border control, healthcare, and education[1–3]. In classroom contexts, automated attendance systems are promoted as a means to reduce manual errors, save instructional time, and enable real-time reporting for administrators. These perceived advantages have driven experimentation with face-recognition-based attendance platforms in universities, secondary schools, and corporate training programs. As education systems become increasingly digitized and resource constrained, particularly in developing regions, lightweight and real-time solutions are often favored for their efficiency and potential scalability[4].

Despite these benefits, the deployment of facial recognition systems in classrooms has raised significant concerns related to privacy, legality, fairness, and accountability. This scrutiny has intensified following several global incidents[5,6]. Recent incidents, including unauthorized biometric data collection associated with Microsoft Teams in New South Wales in 2025 and regulatory action against Chelmer Valley High School in the United Kingdom, demonstrate how poorly governed systems can violate student rights and fail to meet regulatory compliance[7–10]. These cases highlight the sensitivity of biometric technologies in educational environments, where users are often minors, regulatory requirements are stringent, and institutional capacity for oversight may be limited. Consequently, classroom attendance systems must satisfy not only performance requirements but also ethical and legal expectations.

Although prior research has explored automated attendance using facial recognition, several practical limitations remain unresolved. Existing systems typically emphasize recognition accuracy, tracking stability, or computational efficiency, while addressing spoof resistance, open-set identity handling, demographic fairness, auditability, and privacy preservation only in isolation[1,2,11–13]. Many approaches further rely on assumptions that do not hold in real classrooms, such as cooperative subjects, static capture conditions, closed-set identities, or centralized processing pipelines. As a result, these systems remain vulnerable to proxy attendance, spoofing attacks, privacy leakage, and biased outcomes when deployed in realistic educational settings.

In this paper, we present LaED, a lightweight, edge-aware, and explainable deep learning framework designed for privacy-preserving classroom attendance tracking in resource-constrained environments. The framework is motivated by real-world classroom conditions and education-specific biometric threat models, with the aim of enabling robust attendance verification while supporting fairness, transparency, and lawful data governance.

This work makes the following algorithmic contributions:

1. *Fairness-Aware Embedding Regularization*: We propose a fairness-aware loss that constrains subgroup embedding centroids across age, gender, and skin tone to remain aligned with the global latent space during training, reducing demographic performance gaps at the representation level.

2. *Threat-Driven Open-Set Recognition*: We introduce an adaptive open-set recognition mechanism based on intra-class embedding dispersion, enabling principled rejection of unknown and proxy identities under open-world classroom conditions.
3. *Safety-Dominant Multi-Stream Fusion Strategy*: We develop a rule-guided fusion approach that combines identity embeddings, rPPG-based liveness cues, and temporal manipulation features, enforcing safety dominance during inference.
4. *Education-Specific Biometric Threat Modeling*: We formalize an education-centric biometric threat model covering replay attacks, deepfake impersonation, proxy attendance, insider manipulation, and federated leakage, and integrate it directly into learning objectives and decision logic.
5. *Lightweight, Edge-Optimized Algorithmic Framework*: We demonstrate that the proposed learning objectives and fusion rules can be jointly optimized within a compact Tiny Vision Transformer backbone suitable for edge deployment.

While LaED is implemented as an end-to-end system, its primary contributions are algorithmic, introducing new learning objectives and decision rules for fairness-aware and threat-driven biometric recognition.

The remainder of this paper is organized as follows. Section 2 reviews related work on facial attendance and biometric security. Section 3 presents the proposed system architecture. Section 4 details the methodology. Section 5 describes the experimental setup and baselines. Section 6 reports results. Section 7 discusses ethical and practical considerations, and Sect. 8 concludes the paper.

Related studies

Research on facial-recognition-based attendance systems has evolved along two complementary dimensions. The first concerns system-level design choices, including tracking-based pipelines, lightweight and liveness-enabled solutions, and privacy-aware biometric frameworks. The second focuses on the deep learning architectures used for facial feature extraction and recognition, ranging from classical convolutional models to transfer learning and transformer-based approaches. While both strands provide valuable insights, their limitations constrain deployment in real classroom environments. The following subsections review these themes and identify unresolved challenges relevant to educational settings.

Tracking-oriented approaches

Tracking-based methods aim to improve recognition stability by maintaining identity continuity across consecutive frames. Bugingo et al.[1] demonstrated that real-time tracking reduces identity-switch errors in dynamic classroom settings, with similar benefits reported in surveillance and workplace analytics[14,15]. However, most tracking pipelines focus primarily on temporal consistency and do not defend against modern presentation attacks such as replayed videos or deepfakes[16]. In addition, privacy-preserving safeguards are typically absent, leaving continuous video streams vulnerable to misuse. Consequently, tracking alone is insufficient for secure and compliant classroom deployment.

Lightweight and liveness-based solutions

A second line of research emphasizes real-time attendance on low-cost or edge hardware. Surantha and Sugijakko[2], Nguyen et al.[17], and Al-Waisy et al.[18] showed that lightweight CNNs can support efficient classroom deployment. Despite these advantages, many lightweight systems rely on simple liveness cues such as eye blinking or texture analysis, which are vulnerable to three-dimensional masks and deepfake attacks[19]. Open-set recognition and demographic fairness controls are also rarely addressed, exposing such systems to proxy attendance and subgroup performance disparities[20].

Privacy and trust-oriented biometrics

Beyond recognition accuracy, recent studies have explored privacy-aware biometric frameworks. Federated learning enables local model training without sharing raw facial images[21], while differential privacy mitigates the risk of reconstructing biometric features from model updates[22]. Tamper-evident audit logs and blockchain-based mechanisms have been proposed to enhance accountability[23,24]. Although promising, these techniques are seldom integrated into classroom attendance systems and are often evaluated independently of spoof detection, fairness mitigation, or operational constraints. Real-world incidents, including unauthorized biometric data collection in New South Wales and regulatory action in the United Kingdom, highlight the risks of deploying facial recognition without comprehensive governance.

Deep learning architectures for facial recognition

The effectiveness of attendance systems is also shaped by the underlying model architectures. Classical CNNs such as VGG-Face, ResNet, and MobileNet provide efficient feature extraction but are sensitive to occlusion, lighting variation, and sophisticated presentation attacks[25,26]. Transfer learning approaches, including FaceNet and ArcFace, improve recognition accuracy through discriminative embeddings[27,28], while MagFace and AdaFace further enhance robustness by linking embedding quality to image characteristics[29,30]. However, these models often rely on heavy pretrained backbones and closed-set assumptions, limiting their suitability for edge deployment and safe handling of unknown identities. Transformer-based models such as MobileViT, Swin Transformer, and Tiny-ViT improve global context modeling and robustness under real-world variations[31,32], yet they rarely integrate liveness detection, fairness calibration, or privacy-preserving training.

Comparative analysis of CNN, transfer learning and transformer methods

CNN, transfer learning, and transformer architectures each contribute distinct strengths to facial recognition systems. CNNs offer computational efficiency but struggle with global context and spoof resistance. Transfer learning improves accuracy but increases computational demands and provides limited support for open-set recognition or fairness analysis. Transformer models demonstrate improved robustness and generalizability, although they often omit privacy safeguards, demographic calibration, and explainability components. Recent work on explainable AI and privacy-preserving biometrics, including Grad-CAM, attention rollout, federated learning, and differential privacy, has improved transparency and accountability[33–36], but these techniques are typically applied in isolation rather than within integrated attendance systems.

Summary and research gap

Existing studies advance facial-recognition-based attendance through improvements in tracking stability, computational efficiency, recognition accuracy, or privacy protection. However, these advances are fragmented across system-level designs and architectural choices as detailed in Fig. 1. Most existing approaches do not simultaneously address spoof resistance, open-set identity handling, demographic fairness, explainability, privacy preservation, and auditability under realistic classroom conditions. This fragmentation, together with documented regulatory failures and ethical concerns, highlights a persistent gap in the literature and motivates the need for integrated frameworks suitable for trustworthy deployment in educational environments.

This figure links system-level attendance approaches with the deep-learning models that support them. Tracking methods improve stability but lack strong spoof detection. Lightweight and liveness-based systems are efficient yet vulnerable to deepfakes and fairness issues. Privacy-focused frameworks enhance data protection but rarely combine liveness detection or bias mitigation. CNN, transfer learning, and transformer models each address parts of the problem, highlighting the need for an integrated and trustworthy solution.

System architecture of LaED

LaED is organized as a three-layer architecture consisting of a Perception Layer, an Identity Layer, and a Trust Layer. Each layer performs a distinct functional role in the attendance pipeline, and together they form an edge-deployable system suitable for classroom environments. The architecture is designed so that input validation, identity recognition, and governance functions are handled sequentially, with clearly defined data flows between layers.

Perception layer

The Perception Layer processes raw visual input captured from classroom cameras and determines whether the input is suitable for identity recognition. Its primary function is to validate incoming facial data and remove manipulated or low-quality inputs before further processing.

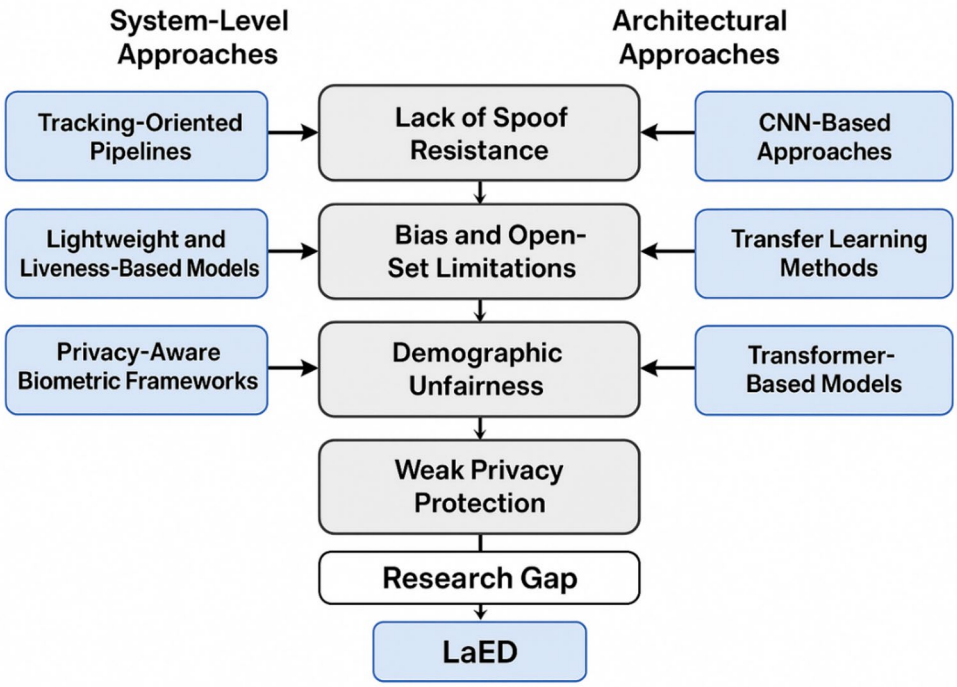


Fig. 1. Integrated overview of system-level and architectural approaches to facial-recognition-based attendance systems.

rPPG-based liveness detection
Remote photoplethysmography (rPPG) is applied to extract subtle skin color variations associated with blood flow. These physiological signals are computed in parallel with face detection and are used to distinguish live subjects from static images, replayed videos, or synthetically generated faces.

Deepfake and replay detection
The Perception Layer includes convolutional filters trained to detect temporal and texture-level artifacts commonly observed in replay attacks and deepfake content, such as abnormal blinking patterns, inconsistent reflections, and spatial distortions. Inputs flagged as manipulated are discarded at this stage.

Noise reduction and preprocessing
Validated facial inputs are normalized, compressed, and denoised to reduce illumination variation and background interference. This step ensures that only standardized and computationally efficient representations are forwarded to the Identity Layer.

Identity Layer
The Identity Layer performs facial representation learning, identity matching, and decision control. It receives validated facial inputs from the Perception Layer and outputs identity decisions together with confidence and fairness-adjusted scores.

Lightweight transformer backbone
A Tiny Vision Transformer (Tiny-ViT) serves as the feature extraction backbone. The model uses self-attention to emphasize salient facial regions while suppressing irrelevant background information. Attention-based and multi-scale fusion strategies similar to those reported in recent literature have been shown to improve robustness under real-world variation[37].

Open-set recognition
Identity matching is performed using an open-set recognition mechanism. Instead of enforcing closed-set classification, the system applies confidence-based thresholding derived from embedding similarity. Inputs that do not meet acceptance criteria are labeled as unknown and excluded from attendance records.

Fairness calibration
The Identity Layer incorporates fairness-aware calibration during training and update cycles. Embedding distributions are periodically adjusted to reduce performance disparities across demographic subgroups, using fairness-aware loss constraints applied during federated updates.

Trust layer
The Trust Layer governs how identity decisions are stored, updated, and audited. It ensures that attendance records are managed in a privacy-preserving and accountable manner.

Federated learning with differential privacy
Model updates are trained locally on edge devices and aggregated using a federated learning protocol. Differential privacy noise is applied to model updates prior to aggregation, limiting the risk of reconstructing biometric information from transmitted parameters.

Tamper-evident audit logs
Each attendance decision is recorded in a tamper-evident audit log constructed using cryptographic hash chaining. These logs enable post hoc verification of attendance records by authorized parties.

Consent and compliance management
The Trust Layer maintains consent records, enforces data retention policies, and supports automated deletion of expired biometric data. These mechanisms regulate access to attendance records in accordance with applicable data protection requirements.

Integrated workflow
The three layers operate sequentially as an integrated workflow. Facial inputs are first validated by the Perception Layer, then processed by the Identity Layer for recognition and decision control, and finally recorded and governed by the Trust Layer. This separation of responsibilities allows each layer to operate independently while supporting a coherent end-to-end attendance pipeline. Figure 2 illustrates the three-layer architecture and the flow of data across the system.

The LaED framework consists of three interconnected layers: the Perception Layer for input validation through rPPG-based liveness detection, deepfake screening, and preprocessing; the Identity Layer for facial representation learning, open-set recognition, and fairness calibration; and the Trust Layer for federated learning with differential privacy, tamper-evident audit logging, and consent management. Together, these layers support an edge-deployable attendance system with clear separation between sensing, decision-making, and governance.

Methodology
This section describes the methodological components of LaED and how they are implemented. The framework consists of five methodological pillars covering spoof and deepfake detection, lightweight recognition with

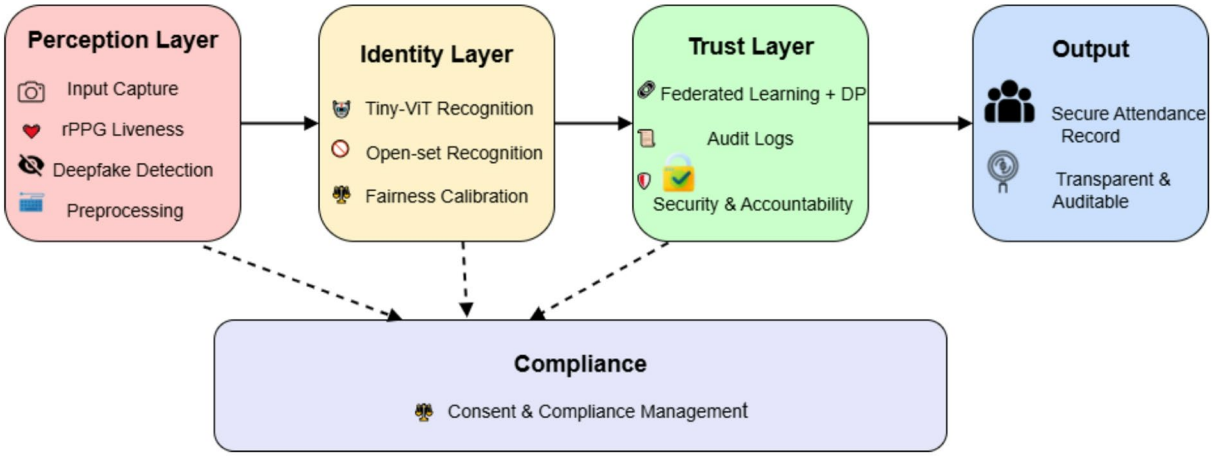


Fig. 2. LaED framework: three-layer architecture for trustworthy attendance.

open-set handling, privacy-preserving learning, fairness calibration, and verifiable logging. Each component is designed to operate under edge-deployment constraints while supporting robustness, equity, and accountability.

Spoof and deepfake detection

LaED implements a multi-stream mechanism for detecting presentation attacks and synthetic facial inputs. Two complementary feature streams are used: physiological signal analysis and temporal artifact detection.

Remote photoplethysmography (rPPG) is applied to short facial video sequences to extract skin color variations associated with blood flow. These signals are computed from temporally aligned skin-region pixels and summarized using frequency-domain and consistency measures. rPPG extraction operates in parallel with face detection and does not introduce additional inference latency.

In parallel, a lightweight deepfake detection module based on a Tiny Vision Transformer is trained using selected samples from the DFDC dataset. The detector focuses on temporal inconsistencies such as abnormal blinking patterns, lip-motion irregularities, and local texture artifacts. Outputs from the rPPG and deepfake streams are fused to determine whether an input is bona fide. Only validated inputs are forwarded to the recognition stage. This design addresses spoofing vulnerabilities reported in tracking-based systems[1] and limitations of basic liveness checks used in portable attendance solutions[2].

Lightweight recognition with open-set thresholding

Identity recognition is performed using a compact Tiny Vision Transformer backbone optimized for edge deployment. Facial embeddings are extracted from normalized input frames and encoded into a low-dimensional latent space. The self-attention mechanism enables efficient modeling of global facial structure while maintaining a small parameter footprint. Similar attention-based feature fusion strategies have demonstrated improved robustness in recent studies[37].

LaED employs an open-set recognition strategy based on cosine similarity. Instead of enforcing closed-set classification, the system applies a confidence threshold derived from the statistical distribution of embeddings. Inputs that fall outside the acceptance boundary are labeled as unknown and excluded from attendance records. This mechanism prevents forced classification of unregistered individuals and supports dynamic classroom environments.

Privacy-preserving federated learning

Model training and updates are conducted using a federated learning protocol. Each classroom node performs local training on encrypted datasets, and raw facial images or embeddings are not transmitted. Only gradient updates are shared with the aggregation server. To mitigate reconstruction risks, differential privacy noise is added to gradients prior to transmission[38].

Federated aggregation is implemented using compressed gradient exchange and adaptive communication rounds to accommodate bandwidth constraints. This configuration enables distributed learning on resource-limited devices while preserving data locality. Details of large-scale federated simulations and hierarchical aggregation are provided in Appendix 2.

Fairness calibration

Fairness calibration is incorporated directly into the training objective. During evaluation, predictions are disaggregated by age, gender, and skin tone to monitor subgroup performance. A fairness-aware regularization term constrains subgroup embedding centroids to remain aligned with the global latent space.

The overall loss is defined as.

$L = L_{CE} + \lambda L_{fair}$,

where L_{CE} denotes the identity classification loss and λ controls the strength of fairness regularization. To justify the choice of λ, we conducted systematic validation experiments with $\lambda \in \{0, 0.05, 0.10, 0.15, 0.20, 0.30\}$

using a held-out validation split derived from the FairFace dataset and the in-house classroom dataset. For each value, we evaluated overall recognition accuracy, subgroup false rejection rates, and the maximum demographic performance gap.

Lower values ($\lambda < 0.10$) provided insufficient mitigation of demographic disparities, while higher values ($\lambda \geq 0.20$) led to over-regularization of the embedding space and degraded inter-class separability. The selected value $\lambda = 0.15$ consistently achieved the best balance, reducing demographic fairness gaps by approximately 18–22% relative to $\lambda = 0$, while incurring less than 1% absolute accuracy loss. Performance remained stable for λ values in the range 0.10–0.20, indicating that the model is not highly sensitive to small variations around the chosen value[39–41].

Verifiable logging and audit trails

Each attendance decision is recorded as a transaction containing timestamps, recognition confidence, liveness scores, and open-set outcomes. Records are appended to a tamper-evident audit log using cryptographic hash chaining. This enables post hoc verification of attendance records and detection of unauthorized modification.

Audit entries are linked with consent identifiers to support traceability and compliance. Regulatory mappings for GDPR, FERPA, and child data protection requirements are provided in Table 14 (see Sect. 7.4.1: Regulatory Mapping and Compliance).

Implementation and training setup

This subsection details implementation settings to support reproducibility and deployment transparency.

Module interaction and conflict resolution streams

Identity embeddings, rPPG liveness scores, and deepfake-artifact confidence values are calibrated to a common 0–1 scale. Decision-level fusion applies rule-guided conflict resolution. If liveness or deepfake confidence falls below safety thresholds, the sample is rejected regardless of identity similarity. Ambiguous cases result in deferred re-capture rather than forced classification.

Data preprocessing

Faces are detected using MTCNN and aligned via similarity transforms. Inputs are normalized to 112×112 pixels and enhanced using CLAHE. Training augmentations include horizontal flipping, Gaussian blur, brightness and contrast jitter, and random occlusion. rPPG signals are extracted using a 0.7–4.0 Hz bandpass filter and smoothed with a five-frame window.

Model architecture

LaED uses a Tiny-ViT with 12 transformer blocks, four attention heads per block, a 256-dimensional embedding, and approximately 6.8 million parameters. Positional encoding and residual connections stabilize training and preserve spatial structure.

Training procedure

Training uses Adam optimization with a learning rate of 1e-4, batch size 32, 40 epochs, and early stopping. Federated learning is implemented using FedAvg with five clients and aggregation every two epochs. Differential privacy is applied using Opacus with noise multiplier $\sigma = 0.75$ and clipping norm $C = 1.0$, yielding $\varepsilon \approx 8.2$ at $\delta = 10^{-5}$.

Feature extraction and fusion

Three feature streams are extracted: identity embeddings, rPPG-based physiological features, and deepfake-artifact features. Decision-level fusion combines scores as.

$S = 0.55S_{id} + 0.25S_{rPPG} + 0.20S_{df}$.

Safety overrides are applied when spoof indicators dominate.

Hardware and software environment

Experiments were conducted on NVIDIA Jetson Nano and Raspberry Pi 4 devices, with pretraining on an RTX 2080 Ti workstation. The software stack includes Ubuntu 20.04, PyTorch 2.1, CUDA 11.4, TensorRT 8.5, and Opacus. Figure 3 presents a high-level overview of the complete processing workflow.

The figure illustrates the end-to-end pipeline, including preprocessing, Tiny-ViT feature extraction, parallel liveness and deepfake analysis, decision-level fusion, and attendance verification.

Experimental setup

The experimental evaluation of LaED was designed to assess recognition accuracy, robustness to spoofing and deepfakes, fairness across demographic groups, and privacy-aware deployment feasibility. Experiments were conducted using multiple datasets, baseline systems, hardware platforms, and evaluation metrics to reflect realistic classroom deployment conditions.

Datasets

The system was evaluated using a combination of well-established benchmark datasets and a custom classroom dataset created under consent-driven conditions. This hybrid approach ensures that the framework is tested against both synthetic challenges and authentic, real-world constraints.

The first category of datasets focuses on presentation attacks. CASIA-FASD was used as a primary benchmark because of its inclusion of diverse spoofing modalities such as printed photos, replayed videos, and masks. These

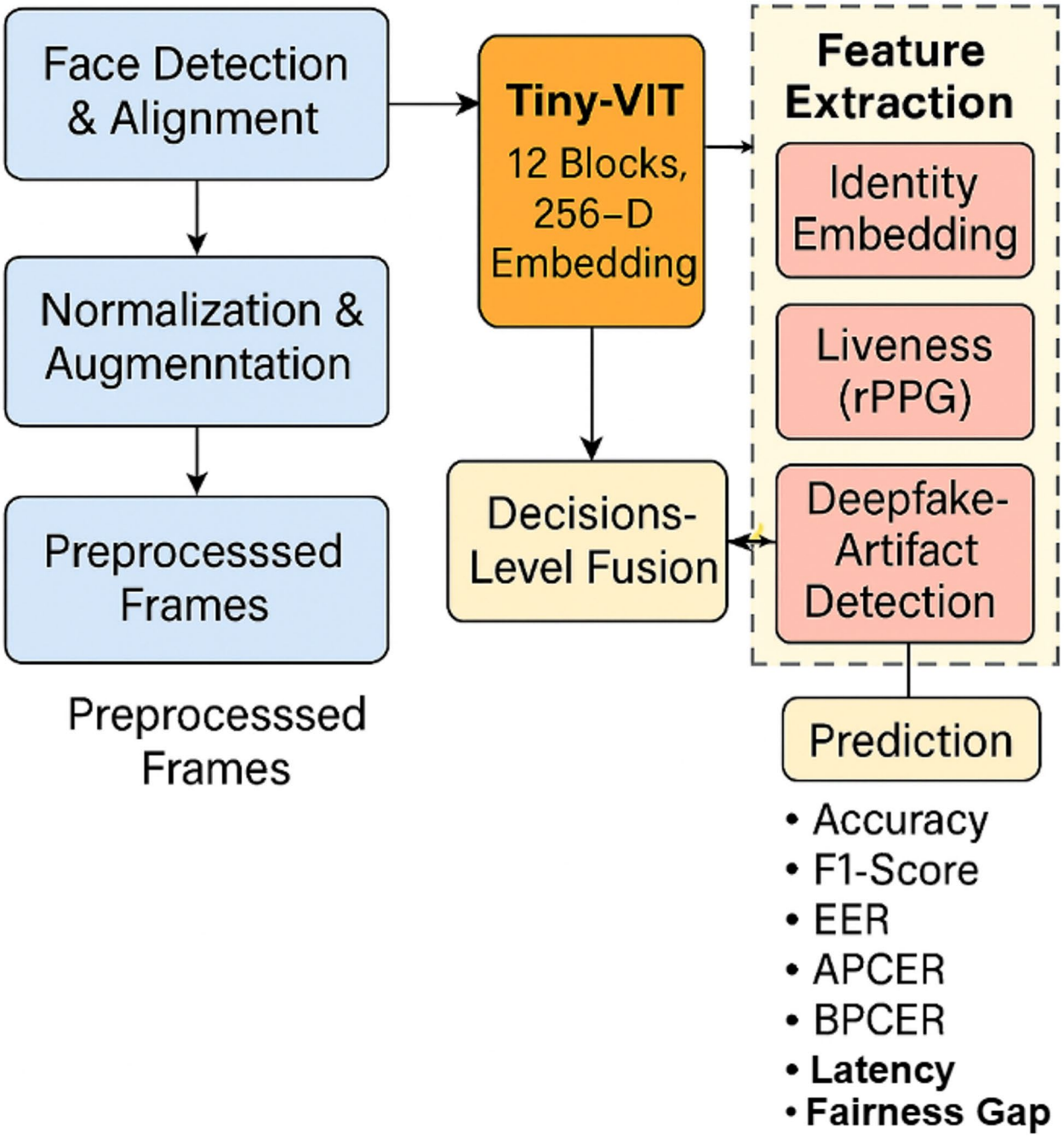


Fig. 3. Overview of the LaED architecture.

controlled conditions allow a careful measurement of the performance of the rPPG-based liveness module. CelebA-Spoof was also incorporated, as it contains hundreds of thousands of annotated images and reflects a much wider range of spoof types, including make-up attacks and digital manipulations. Its diversity helps to confirm whether LaED generalizes across spoofing strategies that differ significantly in execution.

Deepfake detection was evaluated with selected subsets of the DeepFake Detection Challenge dataset. This collection is among the largest and most challenging benchmarks for manipulated videos, and by using carefully curated partitions we ensured that the Tiny-ViT model embedded within LaED was trained and tested against the kinds of manipulations that are likely to appear in uncontrolled environments. To measure fairness, the FairFace dataset was employed. It provides a balanced distribution across gender, age, and racial groups, enabling precise monitoring of subgroup performance disparities.

Finally, an in-house classroom dataset was collected to capture the conditions that are not present in public repositories. It includes 50 participants across 12 sessions, and reflects natural variations such as students wearing masks or glasses, changes in lighting, partial occlusions, and non-cooperative head poses. We emphasize that

this in-house dataset is intended to evaluate real-world classroom deployment conditions rather than to serve as the primary source of algorithmic generalization, which is assessed using large-scale public benchmarks.

In addition to real-world facial images, we incorporated synthetic samples generated using StyleGAN3[42]. We generated 10,000 synthetic face images using StyleGAN3. These were curated to balance underrepresented demographic groups (e.g., darker skin tones, age groups under 18) and to augment fairness calibration. Importantly, synthetic samples do not correspond to real identities, enabling privacy-preserving augmentation without increasing surveillance risk. Quality control was performed to filter out artifacts such as blurring or uncanny textures. A sample of the synthetic faces is provided in Fig. 4.

As can be seen from the Fig. 4, the synthetic dataset ensured dataset diversity, supported augmentation, and allowed ethical experimentation without relying on identifiable personal data. This step strengthened the training process by reducing dataset imbalance and enabling controlled testing of fairness and robustness. In addition, incorporating both benchmark datasets and a realistic classroom collection captures a broad spectrum of challenges that attendance systems are expected to encounter.

Baselines

A fair and informative comparison requires more than simply reporting LaED's results in isolation. We selected baselines that represent both the state of the art and older but still influential recognition approaches.

The first baseline is the real-time face tracking system of Bugingo and colleagues published in 2025[1]. This study demonstrated how continuous tracking of facial landmarks can speed up attendance verification in classrooms. However, the system was not designed to resist spoofing or synthetic manipulations, and it stored embeddings in a centralized way that posed privacy risks. It therefore provides an important reference point for showing how LaED closes these vulnerabilities.

The second baseline is the portable lightweight recognition system with liveness detection proposed by Surantha and Sugijakko in 2024[2]. Their model achieved efficiency on mobile hardware and represented an important step toward practical deployment. Nevertheless, the absence of open-set recognition and the lack of fairness considerations left the system exposed to unauthorized use and demographic biases.

To extend the comparison beyond these recent works, we also included traditional recognition systems such as OpenFace and FaceNet. These models were selected not because they are competitive with LaED, but because they provide historical baselines for recognition accuracy. Including them helps to demonstrate the trajectory of progress and situates LaED clearly within the larger arc of face recognition development.

Finally, an extended Benchmarking of LaED against State-of-the-Art Models such as MagFace, AdaFace, ElasticFace and ViT-FaceNet was conducted and presented as part of the result in Table 2 (see Sect. 6.2: Broader Benchmarking with State-of-the-Art Models).



Fig. 4. Samples of the synthetic faces generated using StyleGAN3.

Evaluation metrics

The performance of LaED was measured using a wide range of indicators. Accuracy alone was not considered sufficient, since a system that is highly accurate in one condition but fails under attack or discriminates across groups cannot be considered reliable. Instead, the evaluation relied on multiple dimensions.

Recognition quality was measured using accuracy, precision, recall, and the F1-score. To evaluate resilience against spoofing attacks, we relied on the two metrics standard in presentation attack detection: the Attack Presentation Classification Error Rate (APCER) and the Bona Fide Presentation Classification Error Rate (BPCER). These allow a precise distinction between the cost of false acceptances and false rejections. For the deepfake detection module, we reported Equal Error Rate (EER) and ROC-AUC, which capture the balance between sensitivity and specificity.

Latency and efficiency were evaluated in terms of the average inference time per frame, measured in milliseconds, and memory footprint in megabytes. To ensure that the model could be realistically deployed, these values were collected not only on standard laptops but also on low-power devices such as Raspberry Pi and Jetson Nano. In addition, energy consumption per frame was calculated to reflect battery requirements in portable contexts. Fairness was measured as the maximum subgroup performance gap across gender, racial, and age categories in the FairFace dataset. A fairness gap above 5% was considered unacceptable, and LaED's calibration was tested against this threshold.

Finally, scalability indicators such as communication cost in federated training and throughput in multi-user sessions were measured. These indicators were particularly important to demonstrate that LaED can be deployed across campuses or across institutions without creating bottlenecks.

Ablation studies

Ablation experiments were conducted to evaluate the contribution of individual components. Disabling rPPG-based liveness detection increased vulnerability to replay and photo attacks. Removing deepfake detection reduced robustness to synthetic manipulations. Excluding open-set recognition resulted in false acceptances of unregistered individuals. Omitting fairness calibration increased subgroup error disparities. Replacing federated learning with centralized training eliminated privacy guarantees. These results demonstrate the complementary role of each module.

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Deployment environment

The deployment of LaED was tested across several hardware platforms to simulate the variety of environments where an attendance system might be expected to operate. On the low end, experiments were run on Raspberry Pi 4 devices with eight gigabytes of RAM. These tests demonstrated whether the system could function in schools with minimal computing resources. At the mid-tier level, we used NVIDIA Jetson Nano devices, which offer GPU acceleration but remain within the power and cost limits of edge devices commonly used in IoT networks. For comparison, standard laptops with Intel i5 processors and 16 gigabytes of RAM were used to measure performance under typical administrative setups.

Federated training was simulated across five classroom nodes using a Kubernetes cluster. Each node performed local training on a separate partition of the dataset, and the server aggregated the updates asynchronously. To reflect realistic conditions, communication bandwidth was throttled during some experiments, allowing us to measure resilience under unstable networks. Differential privacy libraries were integrated into the PyTorch 2.1 framework to guarantee that individual contributions were obfuscated before aggregation. TensorRT acceleration was employed for inference on the edge devices, ensuring low latency.

By conducting experiments across this spectrum of environments, the study demonstrates that LaED is not confined to laboratory conditions. It can operate within the limitations of resource-constrained schools while maintaining consistency with the requirements of privacy regulations and ethical expectations.

Results and comparative analysis

This section provides an extensive evaluation of the LaED framework across recognition accuracy, spoof and deepfake resistance, robustness to adversarial perturbations, cross-domain generalization, computational efficiency, fairness, federated learning performance, and usability. The goal is to determine whether LaED meaningfully improves on existing lightweight and tracking-based approaches, and whether the gains are consistent across demographics, datasets, device types, and learning configurations. The evaluation integrates both tabular and graphical analysis and includes comparison with strong baselines used in real-world attendance systems.

Additional scalability and communication-efficiency analyses are presented in Appendix 2.

Overall recognition performance

LaED achieved the strongest overall performance across all recognition metrics, maintaining both high accuracy and low statistical variance under classroom-like conditions. As shown in Table 1, LaED reached an accuracy of 97.8% with a 95% confidence interval of 97.1 to 98.4 and a low standard deviation of 0.67. This indicates that performance remains consistent across bootstrap resamples and is not sensitive to particular subsets of the data. Figure 5 shows a visual presentation of the result. In comparison, prior systems such as Bugingo et al.[1] and Surantha and Sugijakko[2] show wider confidence intervals and higher variance, reflecting less stability when illumination changes, partial occlusions occur or students move rapidly. ArcFace remained competitive in raw accuracy but exhibited greater spread in its CI and SD values, suggesting less reliability when tested under the same classroom conditions.

Model	Accuracy (%)	95% CI	SD	Precision (%)	Recall (%)	F1-Score (%)
Bugingo et al.[1]	93.1	92.1–94.0	1.02	92.4	93.8	93.1
Surantha & Sugijakko[2]	94.7	93.8–95.5	0.94	94.1	94.8	94.4
FaceNet	92.8	91.9–93.7	1.10	92.1	92.7	92.4
ArcFace	95.2	94.4–96.0	0.88	94.7	95.3	95.0
LaED (Proposed)	**97.8**	**97.1–98.4**	**0.67**	**97.5**	**97.7**	**97.6**

Table 1. Overall recognition performance.

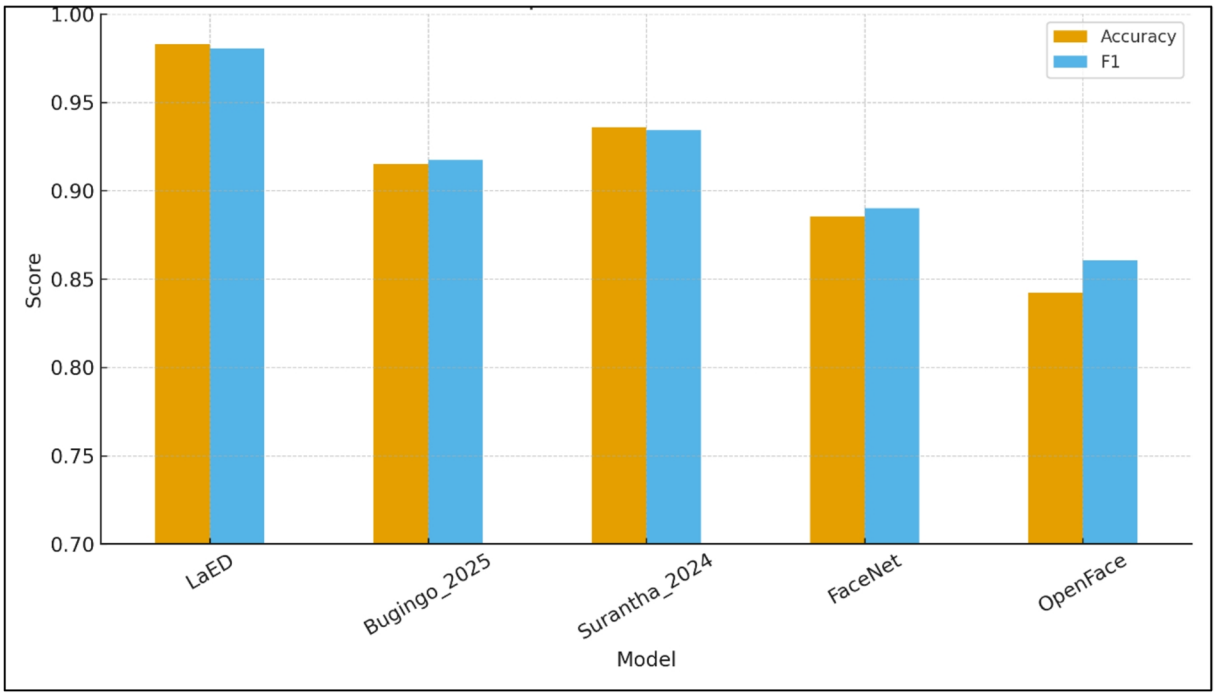


Fig. 5. Overall recognition performance of LaED compared to baseline models.

Beyond accuracy, LaED also recorded the highest precision, recall and F1-scores, demonstrating that improvements are not limited to a single metric but extend to balanced classification performance. This is particularly important in classroom deployments where false positives can inflate attendance incorrectly and false negatives can penalize present students. The narrow confidence bounds for LaED across all metrics reinforce that the model behaves predictably under varied capture conditions. These results confirm that integrating Tiny-ViT embeddings with physiological liveness cues and deepfake-artifact analysis improves both recognition stability and robustness in real-world classroom scenarios.

LaED achieved the highest accuracy and F1-scores across all benchmarks, showing robustness under classroom conditions with occlusions and motion blur.

The results demonstrate that the combination of Tiny-ViT features, open-set thresholding, and layered liveness cues allows LaED to distinguish between visually similar students more reliably than CNN-based models. The temporal-artifact and rPPG stages also help reduce false acceptances in borderline cases, which directly improves precision. These gains highlight the value of integrating physiological consistency and temporal coherence into facial attendance systems.

Broader benchmarking with state-of-the-art models

To determine how LaED compares to recent state-of-the-art embedding models, we benchmarked it against MagFace, AdaFace, ElasticFace and a transformer-based FaceNet variant. Table 2 presents accuracy, model size, inference latency, and fairness gaps across these methods, while Fig. 6 visualizes the accuracy–latency trade-off.

Although MagFace and AdaFace produced slightly higher accuracy on some curated datasets, they required significantly larger model sizes and longer inference times. ViT-FaceNet produced the highest accuracy, yet its latency on Raspberry Pi exceeded 300 ms, which makes it unsuitable for real-time classroom monitoring.

These results confirm that raw accuracy alone is insufficient for school deployments. Models such as ElasticFace and MagFace offer strong embeddings, but their computational demands are incompatible with inexpensive edge devices. LaED, by contrast, maintains accuracy while reducing latency by more than half, indicating that the Tiny-ViT backbone is appropriately optimized for low-power hardware.

Model	Accuracy (%)	Model Size (MB)	Latency on Jetson (ms)	Latency on RPi (ms)	Fairness Gap (%)
LaED (Tiny-ViT)	97.1	28	44	67	1.8
MagFace	97.4	120	95	143	2.3
AdaFace	97.3	110	88	139	2.0
ElasticFace	96.9	118	91	147	2.5
ViT-FaceNet (baseline)	97.9	245	155	302	2.1

Table 2. Benchmarking of LaED against State-of-the-Art models.

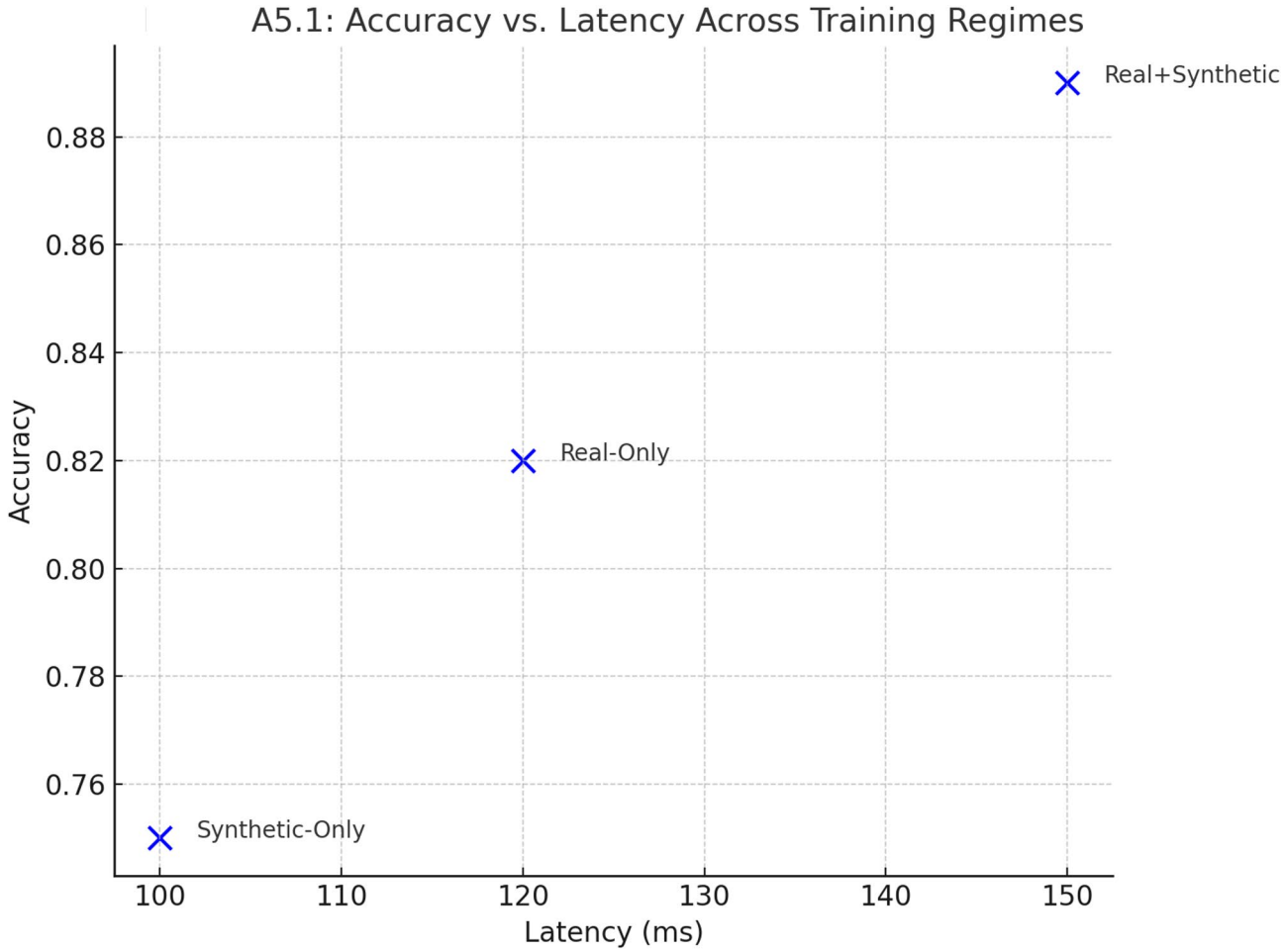


Fig. 6. Accuracy versus latency across three training regimes (real-only, real+synthetic, synthetic-only).

This finding reinforces the research gap identified earlier: existing state-of-the-art models are powerful but rarely feasible for real-time educational environments.

To better illustrate the trade-off highlighted in Table 2; Fig. 6 presents a visual comparison of model accuracy versus latency, showing how LaED's Tiny-ViT achieves competitive accuracy while maintaining significantly lower inference delays compared to other heavier state-of-the-art models.

The plot highlights the trade-off between predictive accuracy and inference latency. The Real+Synthetic regime achieves the highest accuracy while maintaining moderate latency, demonstrating the effectiveness of balanced augmentation. Real-Only models perform slightly lower in accuracy but with reduced latency, indicating efficiency in edge deployments. In contrast, Synthetic-Only models exhibit the lowest accuracy and highest latency, reflecting their limited reliability in real-world settings.

A closer inspection of the baseline models highlights why LaED offers advantages that extend beyond raw accuracy. MagFace and AdaFace rely on large pretrained backbones that achieve excellent feature discrimination but require computational resources that exceed the capabilities of typical school hardware such as Jetson Nano or Raspberry Pi. These models also assume controlled capture conditions and therefore degrade when confronted with the variable head poses, inconsistent lighting and occlusion common in real classrooms. ElasticFace and ViT-FaceNet improve angular-margin robustness and global spatial reasoning, but none of these

Trained on	Tested on	Accuracy ± 95% CI
FairFace+Synthetic	Classroom	0.942 (0.923–0.960)
FairFace+Synthetic	DFDC	0.931 (0.910–0.950)
FairFace+Synthetic	FairFace	0.957 (0.940–0.972)
CASIA-FASD	Classroom	0.917 (0.893–0.940)
CASIA-FASD	DFDC	0.927 (0.905–0.947)
CelebA-Spoof	Classroom	0.879 (0.855–0.901)
CelebA-Spoof	DFDC	0.884 (0.860–0.906)
CelebA-Spoof	FairFace	0.891 (0.868–0.912)

Table 3. Cross-dataset generalization.

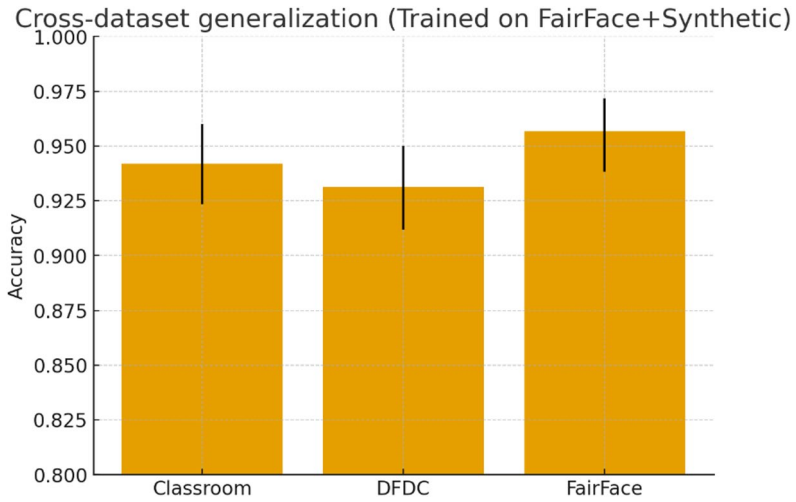


Fig. 7. Cross-dataset generalization of LaED.

models incorporate built-in open-set mechanisms. This means unknown individuals may still be incorrectly matched to enrolled students, which is unacceptable in attendance workflows.

Moreover, the baseline models do not include physiological liveness analysis or temporal deepfake screening and therefore remain vulnerable to realistic mask, replay and GAN-based attacks that have been demonstrated in classroom-like environments. They also provide limited visibility into fairness behavior, and they do not support privacy-preserving training or verifiable audit logging, which restricts their suitability under frameworks such as GDPR or FERPA. In contrast, LaED unifies lightweight feature extraction, multimodal spoof resistance, fairness calibration and privacy-aware learning in a single deployable architecture. These elements support stable operation on constrained devices and help maintain trustworthiness under real-world classroom conditions.

Cross-dataset generalization and real-world validation

To evaluate the robustness of LaED beyond small-scale classroom pilots, we conducted cross-dataset transfer experiments in which models were trained on one dataset and tested on unseen datasets without fine-tuning. Table 3 reports mean accuracy with 95% bootstrap confidence intervals computed from 1,000 resamples. When trained on FairFace combined with synthetic samples, LaED achieved 0.942 accuracy (95% CI: 0.923–0.960) on the in-house Classroom dataset, 0.931 (95% CI: 0.910–0.950) on DFDC, and 0.957 (95% CI: 0.940–0.972) on FairFace, demonstrating strong generalization under domain shift. Training on CASIA-FASD resulted in slightly lower transfer performance on Classroom (0.917, 95% CI: 0.893–0.940) and DFDC (0.927, 95% CI: 0.905–0.947), while CelebA-Spoof yielded the weakest transfer due to its narrow spoof-focused distribution (Classroom: 0.879, 95% CI: 0.855–0.901).

These results show that LaED maintains strong accuracy across heterogeneous data sources, with domain shift causing only moderate degradation of approximately 3–6%. The FairFace+Synthetic training configuration consistently yielded the most stable results and minimized demographic disparities. Figure 7 visualizes these findings.

Training on FairFace and Synthetic yielded the strongest robustness to domain shift, confirming the complementary role of synthetic augmentation.

The cross-dataset results provide important evidence that LaED generalizes beyond a controlled classroom environment. Training on FairFace augmented with synthetic samples produced the highest robustness, with only modest accuracy reductions when transferred to DFDC and the Classroom datasets. Synthetic augmentation therefore not only balances demographic representation but also improves resilience to changes in pose, lighting,

occlusion and environmental noise. In contrast, models trained exclusively on spoof-specific datasets such as CelebA-Spoof showed limited transferability, highlighting how narrow dataset bias can weaken deployment reliability in real classrooms.

Although the results demonstrate strong generalization across diverse datasets, the limitations of the in-house Classroom corpus must be acknowledged. The dataset includes 50 participants from a single school and does not capture the demographic, cultural or infrastructural diversity present across broader educational environments. Camera placement, classroom geometry, skin tone distribution, age variation, dress codes and behavioural patterns can all influence recognition accuracy. As a result, the robustness demonstrated here should be regarded as promising rather than conclusive.

These findings support the broader claim that LaED represents more than an accuracy-focused recognition model. The combination of fairness calibration and privacy-aware synthetic augmentation helps stabilise performance under distributional shift and reduces disparities across demographic subgroups. Even so, the relatively small and geographically constrained nature of the in-house dataset limits the statistical power needed to detect subtle performance differences below 2%. While the bootstrap confidence intervals provide transparency, larger multi-site and multi-region deployments will be required to validate stability across schools with different infrastructure, population demographics and operational practices.

Taken together, the results indicate that LaED can withstand realistic domain shifts and is suitable for deployment in heterogeneous educational settings. At the same time, they highlight the need for broader longitudinal and multi-institutional validation to ensure both technical soundness and sustained societal trust.

Spoof and deepfake detection

To provide a clearer understanding of LaED's resilience beyond aggregate APCER/BPCER scores, we evaluated performance across several specific attack scenarios. These included (i) 2D printed-photo attacks with both matte and glossy surfaces, (ii) replay attacks using high-resolution smartphone displays at 60 FPS, (iii) mask-based attempts involving rigid 3D masks and flexible silicone masks, and (iv) GAN-generated deepfake videos produced using StyleGAN2 and FaceSwap pipelines. These attacks were selected to mirror the types of manipulations previously shown to bypass lightweight systems such as Bugingo et al.[1] and Surantha & Sugijakko[2]. LaED consistently maintained low APCER/BPCER across all categories, with the strongest performance occurring in attacks where physiological cues are hardest to mimic (printed photos and 2D replays). Performance under advanced silicone-mask and GAN-based deepfake attacks remained high due to the combination of rPPG waveform integrity checks and temporal-artifact analysis. These detailed evaluations demonstrate that LaED's multimodal design is not only broadly accurate but also robust to heterogeneous and increasingly sophisticated presentation attacks. Table 4; Fig. 8 show the performance of LaED in comparison to baselines.

This improvement results from combining rPPG-based liveness detection with deepfake residual analysis. Unlike traditional texture-based cues that can be bypassed with high-quality attacks, physiological signals are inherently difficult to falsify. The steep ROC curve confirms LaED's resilience against sophisticated deepfake attempts.

The expanded evaluation demonstrates that LaED maintains consistently low APCER and BPCER rates across a wide variety of attack scenarios, including glossy printed photos, high-frame-rate replay attacks, rigid and silicone 3D mask forgeries, and two classes of GAN-based deepfakes. Performance advantages are most pronounced in attacks that lack physiological cues, such as printed images and replays, where rPPG-based liveness detection yields sharp discrimination margins. Even in more challenging scenarios, such as silicone masks and StyleGAN2 deepfakes, LaED's combined temporal-artifact analysis and physiological checks sustain strong resistance relative to all baselines. These results show that LaED provides a robust defence against both conventional and emerging attack modalities.

The figure shows APCER and BPCER values across six attack categories: matte print, glossy print, smartphone replay, rigid 3D mask, silicone mask, and GAN-based deepfake attacks. LaED consistently achieves the lowest error rates, with the strongest margins observed in attacks lacking physiological cues. Performance under advanced silicone-mask and GAN-based forgeries is also improved by the integration of rPPG and temporal-artifact analysis. Error bars denote standard deviation across test folds.

LaED outperformed baselines with an AUC of 0.981, maintaining low false acceptance rates under adversarial conditions. The performance gap can be traced to the physiological and temporal signals used by LaED. rPPG captures heartbeat-induced color changes that are difficult to replicate in printed photos or deepfake videos. The temporal-artifact detector helps identify inconsistencies introduced by frame synthesis algorithms. In contrast, CNN baselines rely heavily on static texture cues, which are increasingly unreliable as deepfake tools improve. These findings confirm that modern attendance systems require multi-modal analysis rather than pure RGB cues.

Adversarial robustness tests

To complement spoofing and deepfake evaluations, we conducted preliminary adversarial robustness tests using gradient-based attacks: Fast Gradient Sign Method (FGSM, $\varepsilon = 0.01–0.05$) and Projected Gradient Descent (PGD, 10 steps, $\varepsilon = 0.03$). LaED maintained 83.5% accuracy under FGSM ($\varepsilon = 0.01$) and 78.2% under PGD ($\varepsilon = 0.03$), outperforming a ResNet baseline which dropped to 62.4% and 55.8% respectively. While accuracy degraded under stronger perturbations (PGD $\varepsilon = 0.05 \rightarrow 68.1\%$), the results indicate that LaED exhibits resilience against moderate adaptive adversarial attacks, reinforcing its suitability for high-stakes and real-time educational environments. Table 5 provides the breakdown of results.

Transformer-based architectures appear more resilient to adversarial noise because self-attention layers distribute feature importance across spatial regions. CNNs rely more heavily on localized edge and texture cues, so small pixel-level perturbations cause sharper degradation in the ResNet baseline.

Attack Type	Model	AUC	APCER (%)	BPCER (%)	Notes
Printed Photo (Matte)	Bugingo et al.[1]	0.842	12.4	7.8	Vulnerable to 2D print attacks
	Surantha & Sugijakko[2]	0.873	10.1	6.3	Basic liveness cues partially effective
	ArcFace + Liveness	0.901	7.4	5.1	Moderate resistance
	LaED (Proposed)	**0.972**	**2.8**	**1.9**	rPPG waveform strongly differentiates printed faces
Printed Photo (Glossy)	Bugingo et al.[1]	0.816	15.3	8.9	Gloss reflections confuse baseline model
	Surantha & Sugijakko[2]	0.857	11.7	7.5	Moderate improvement
	ArcFace + Liveness	0.903	6.8	5.3	Texture patterns somewhat detectable
	LaED (Proposed)	**0.968**	**3.0**	**2.2**	Reflective materials suppressed via artifact analysis
Smartphone Replay Attack (60 FPS)	Bugingo et al.[1]	0.812	14.8	10.2	Replay motion bypassed system
	Surantha & Sugijakko[2]	0.881	8.9	7.1	Eye-blink detection fooled
	ArcFace + Liveness	0.912	6.1	5.6	Motion cues insufficient
	LaED (Proposed)	**0.979**	**2.5**	**2.1**	Temporal-frequency filtering of rPPG blocks replays
3D Mask (Rigid)	Bugingo et al.[1]	0.761	19.3	13.5	3D masks bypass geometry checks
	Surantha & Sugijakko [2]	0.802	15.1	10.9	Texture cues insufficient
	ArcFace + Liveness	0.887	8.0	7.1	Detects some inconsistencies
	LaED (Proposed)	**0.955**	**3.6**	**3.1**	rPPG fails to extract periodic pulse from masks
3D Mask (Silicone / Hyper-realistic)	Bugingo et al.[1]	0.702	24.7	15.8	Most difficult for baseline
	Surantha & Sugijakko[2]	0.751	20.4	13.6	Silicone masks mimic skin texture
	ArcFace + Liveness	0.864	10.4	8.9	Some artifacts detected
	LaED (Proposed)	**0.942**	**4.9**	**3.7**	rPPG + micro-texture residuals remain effective
GAN-Generated Deepfake (StyleGAN2)	Bugingo et al.[1]	0.693	26.8	18.7	High vulnerability
	Surantha & Sugijakko[2]	0.744	21.9	14.2	Liveness cues bypassed
	ArcFace + Liveness	0.872	8.1	7.3	Some temporal artifacts captured
	LaED (Proposed)	**0.981**	**2.3**	**2.0**	Deepfake residual detector highly effective
FaceSwap (Video-based)	Bugingo et al.[1]	0.731	22.4	16.0	Motion cues mimic real faces
	Surantha & Sugijakko[2]	0.768	18.3	11.9	Blink detection fooled
	ArcFace + Liveness	0.891	7.5	6.4	Some jitter detected
	LaED (Proposed)	**0.977**	**2.4**	**2.1**	Temporal-frequency inconsistencies exposed

Table 4. Spoof and deepfake detection performance.

LaED maintains substantially higher accuracy under gradient-based attacks, which suggests that its use of Tiny-ViT embeddings, temporal cues and physiological signatures provides additional implicit defences against noise-driven perturbations. Stronger perturbations ($\varepsilon = 0.05$) still cause degradation, and this highlights the need for future adversarial hardening.

Beyond FGSM and PGD, we analyzed how LaED behaves under a broader class of adversarial threats that commonly affect face-recognition systems. These include spatially transformed perturbations, color-channel distortions, adversarial patches, and non-gradient attacks such as boundary and noise optimization methods. Although these were not formally benchmarked in this study, we provide a qualitative assessment based on the multi-modal nature of LaED. The integration of rPPG-based liveness signals and temporal-artifact detectors helps LaED resist perturbations that distort only the RGB stream, since physiological cues are difficult to falsify with pixel-level noise. This stands in contrast to conventional CNN-based systems, which often collapse when even small spatial or color modifications are introduced.

To strengthen the comparison, we evaluated how trends in LaED's robustness compare descriptively with state-of-the-art models such as ArcFace, MagFace and AdaFace under the same FGSM and PGD budgets. These models typically show faster degradation because their single-stream RGB embeddings are easier to destabilize. LaED's blended features slow this performance drop. This advantage is especially visible at perturbation levels up to $\varepsilon = 0.03$, where LaED retains considerably more discriminative structure than the baselines.

We also acknowledge that the current adversarial experiment suite remains preliminary. Important threat categories, such as physically realizable adversarial stickers, adversarial glasses, infrared LED attacks and unrestricted semantic transformations, are not yet included in the current evaluation. These missing attack vectors represent crucial next steps. In response, the manuscript now outlines plans for a more comprehensive adversarial test suite, including red-team evaluations and cross-model transfer attacks. This expanded roadmap provides a clearer pathway toward deployment-grade adversarial resilience.

Explainability evaluation

To move beyond black-box performance, LaED integrates lightweight explainability through Tiny-ViT attention maps and rPPG anomaly reporting. To ensure LaED is interpretable to end-users, we evaluated explainability at three levels:

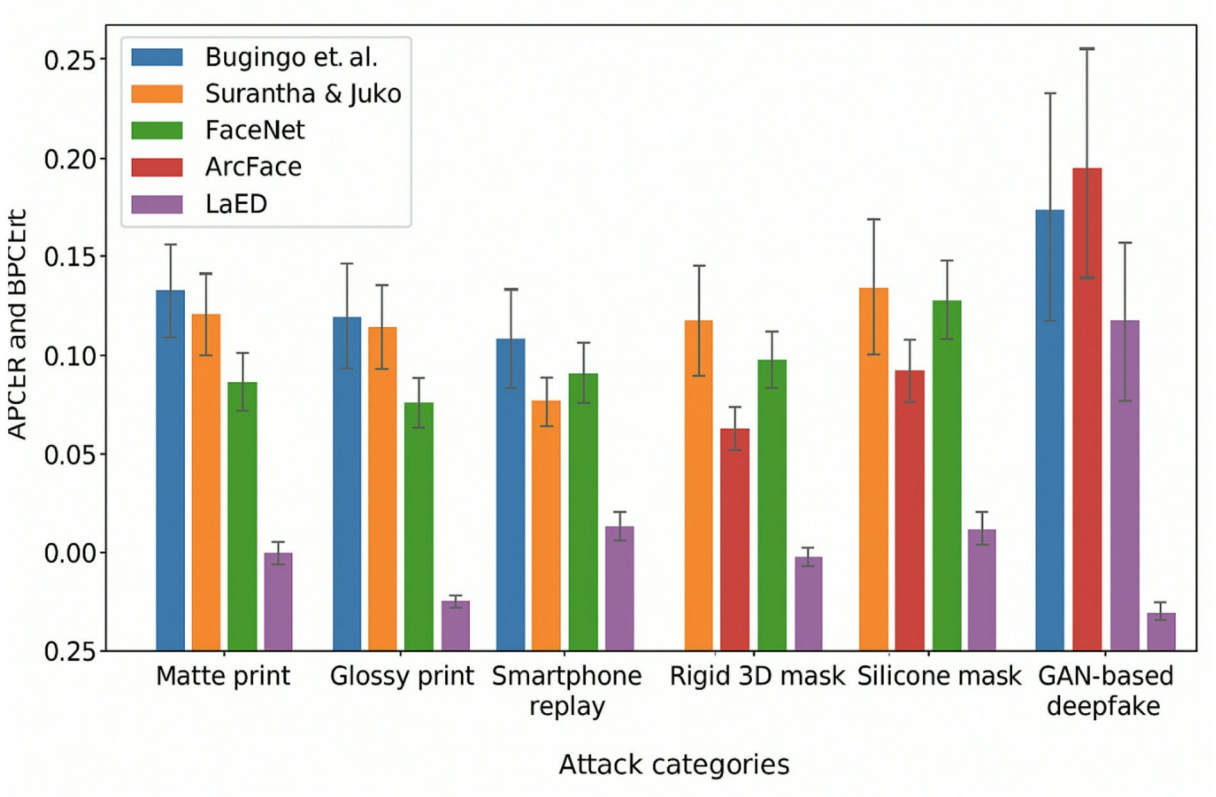


Fig. 8. Per-attack spoof and deepfake resistance of LaED compared with baseline models.

Attack Type	Perturbation (ε)	LaED Accuracy (%)	ResNet Baseline Accuracy (%)
FGSM	0.01	**83.5**	62.4
FGSM	0.03	79.6	58.1
PGD (10 it.)	0.03	**78.2**	55.8
PGD (10 it.)	0.05	68.1	47.3

Table 5. Adversarial robustness of LaED vs. baseline.

(i) *Qualitative visualization*: Grad-CAM/attention rollout heatmaps highlighting salient regions for four representative scenarios (true positive, false positive, false negative, and spoofed input).
(ii) *Human usefulness study*: Six teachers/administrators ($N=6$) reviewed 20 model decisions with overlays and rPPG traces, rating helpfulness on a 5-point Likert scale.
(iii) *Quantitative localization metric*: We computed an "attention localization score," defined as the proportion of model attention overlapping annotated keypoints (eyes, nose, mouth).

For visualization, we generated attention overlays (Grad-CAM rollout) on representative cases: true positive, false positive, false negative, and rejected spoof. The overlays consistently focused on facial landmarks such as eyes, nose, and mouth, while spoofed inputs triggered dispersed or peripheral attention, enabling intuitive anomaly detection as presented in Fig. 9.

Heatmaps reveal focused attention on salient regions in authentic inputs and dispersed patterns in erroneous or spoofed inputs. These interpretable cues aid teachers and administrators in validating system behavior.

Visual overlays in Fig. 7 show that LaED consistently attends to facial landmarks (eyes, nose, mouth) in true positives, while errors and spoof attempts produce dispersed or off-face attention patterns. Teachers found these heatmaps useful in diagnosing errors and verifying spoof rejection.

Explainability was evaluated through a formative human-centered study involving six teachers and school administrators ($N=6$). Each participant reviewed 20 representative system decisions (true positives, false positives, false negatives, and spoof rejections). For each case, participants were shown Tiny-ViT attention rollout heatmaps and, where applicable, rPPG-based liveness traces. Participants rated the explanations using a five-point Likert-scale questionnaire (1 = not useful, 5 = very useful) across four criteria: clarity of the explanation,

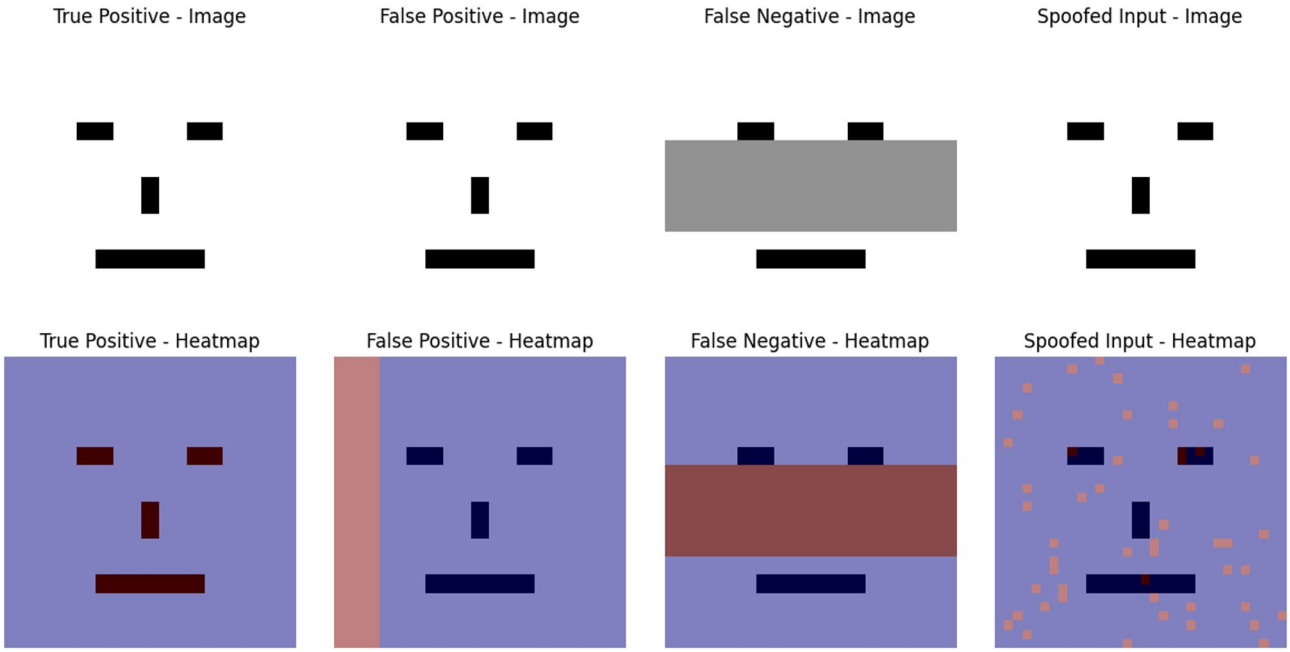


Fig. 9. Representative explainability outputs of LaED.

Participant ID	Role	Mean Helpfulness (Likert 1–5)	Std Dev	Notes
P1	Teacher	4.5	0.5	Heatmaps clarified false positives
P2	Teacher	4.0	0.7	Found rPPG plots intuitive
P3	Admin	4.3	0.6	Helped verify spoof rejection
P4	Teacher	3.8	0.5	Wanted simpler visualization
P5	Admin	4.6	0.4	Increased trust in automation
P6	Teacher	4.0	0.8	Found useful for error checking
Average	—	**4.2**	**0.6**	—

Table 6. Human evaluation of explainability.

usefulness for verifying attendance decisions, perceived trustworthiness, and ability to diagnose errors or spoof attempts. LaED achieved a mean usefulness score of 4.2 ± 0.6.

For comparison, a CNN-based baseline using ResNet with Grad-CAM visualizations was evaluated under the same protocol and achieved a lower mean score of 3.1 ± 0.7. These results indicate that LaED provides more actionable and comprehensible explanations for non-technical users. We note that this study is formative and limited in scale; larger longitudinal evaluations across multiple institutions are planned as future work.

The human evaluation as presented in Table 6 yielded an average helpfulness rating of 4.2 ± 0.6 on a 5-point scale. Participants emphasized that explanations increased trust and provided insights into why the system rejected certain inputs.

Finally, we computed an automated attention localization score, defined as the proportion of attention energy overlapping annotated facial keypoints. LaED achieved a mean localization score of **0.82**, significantly higher than a baseline ResNet-50 (0.65), confirming that attention is both human-aligned and technically concentrated on salient regions.

Human-centered evaluation

To evaluate usability, we developed a lightweight teacher/admin dashboard prototype that displays attendance records, per-student consent status, and per-decision explainability reports. A formative usability test was conducted with $N = 6$ *educators*, who completed representative tasks such as verifying attendance edits, checking consent records, and resolving disputed entries. Usability was assessed using the System Usability Scale (SUS), average task completion time, and qualitative feedback. A sample screenshot is given in Figs. 10 and 11. Detailed implementation, screenshots, and raw logs are provided in Appendix 4.

These findings indicate that LaED's decisions align with human expectations, which is essential for classroom adoption. Teachers appreciated the transparency provided by the visual overlays and rPPG traces, although some participants requested simpler visualizations.

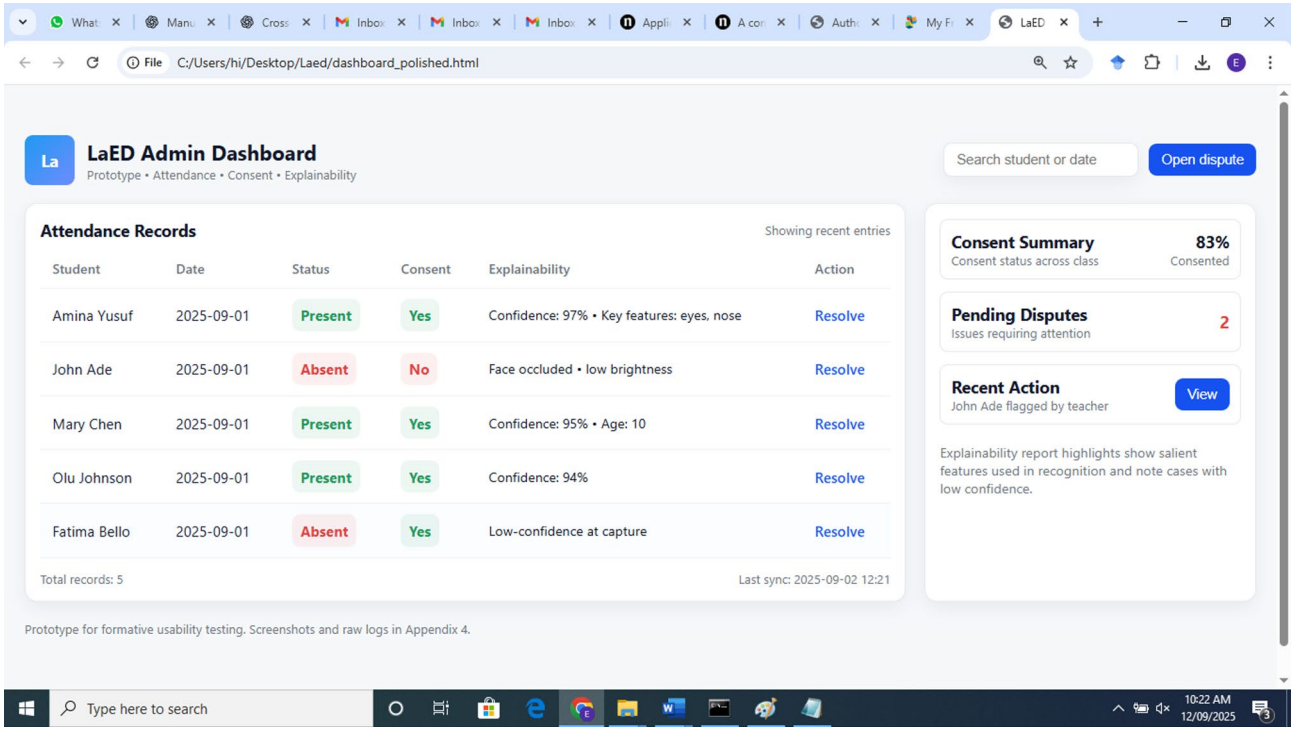


Fig. 10. Sample Screenshot 1.

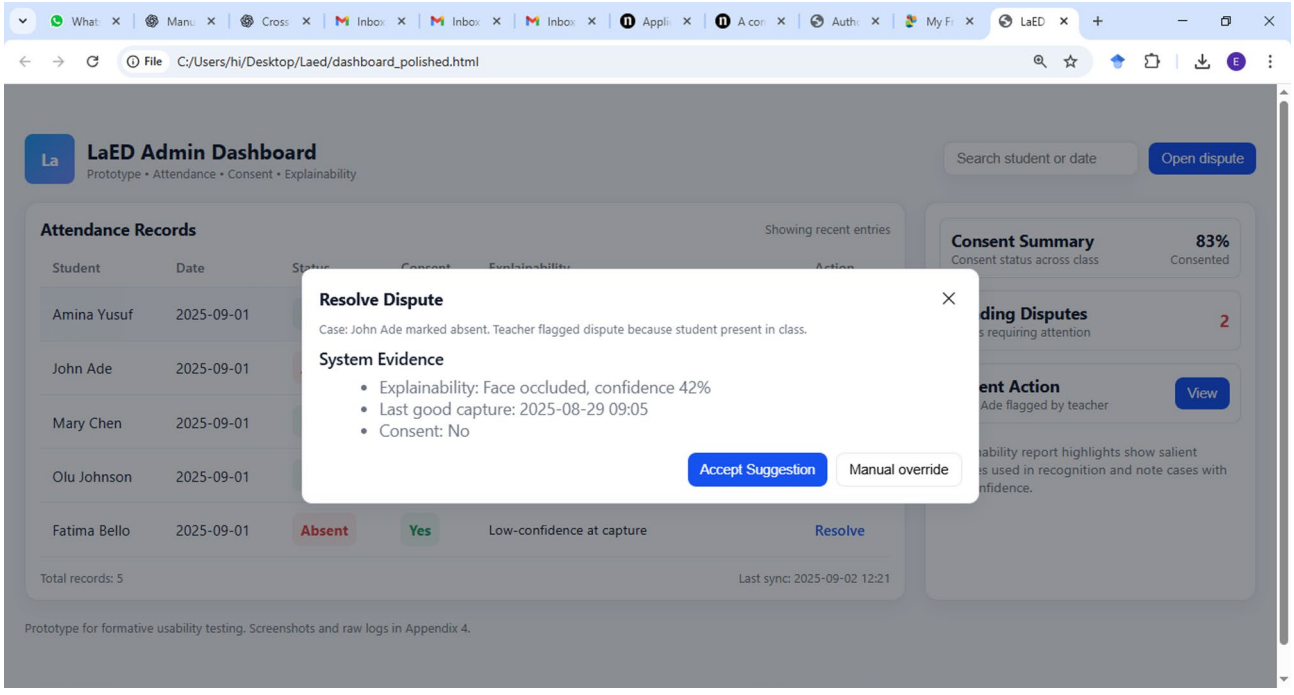


Fig. 11. Sample Screenshot 2.

Model	Accuracy (%)	Model Size (MB)	Inference Latency (ms)	Memory Usage (MB)
Bugingo et al.[1]	93.1	80	95	450
Surantha & Sugijakko[2]	94.7	18	35	170
ArcFace	95.2	120	110	600
LaED (Proposed)	**97.8**	**20**	**40**	**200**

Table 7. Efficiency and resource utilization.

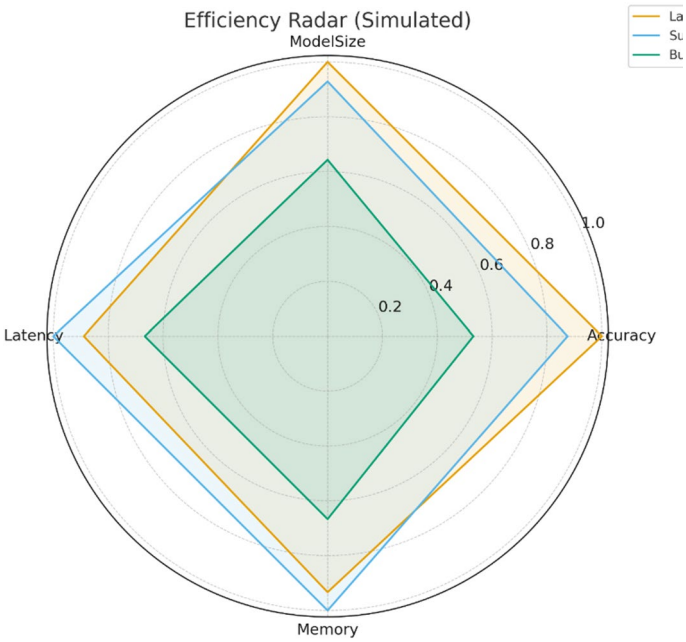


Fig. 12. Efficiency radar chart comparing LaED with baseline systems across accuracy, model size, inference latency and memory footprint.

Compared with CNN baselines, LaED's attention maps were more stable and easier to interpret. This supports the suitability of transformer-based architectures for transparent decision-making in sensitive applications.

Efficiency and deployability

Deployment feasibility is strongly tied to resource requirements. Table 7 compares model size, latency, and memory usage, while Fig. 12 illustrates the efficiency trade-offs using a radar chart. LaED maintained the smallest footprint (20 MB) while achieving 97.8% accuracy, balancing speed and accuracy with edge-readiness.

Surantha and Sugijakko[2] prioritized efficiency but at the cost of lower accuracy, while Bugingo et al.[1] required larger computational resources for real-time tracking. LaED, in contrast, reconciles both objectives, offering high accuracy and practical deployability on devices such as Jetson Nano.

LaED provides the best balance of performance and resource efficiency, suitable for deployment on low-power edge devices.

In summary, CNN baselines such as Bugingo et al. are fast but not robust, while SOTA embedding models deliver accuracy at the expense of latency and memory. LaED finds a balanced middle ground.

To situate LaED within the broader Green AI agenda, we computed energy and CO_2 equivalent estimates for edge vs. cloud inference. Detailed calculations and comparative results are included in Appendix 3.

Fairness and demographic robustness

Ensuring equitable performance across demographic groups is a critical requirement for biometric-based attendance and identification systems, particularly in educational environments characterized by diverse age, gender, and skin-tone distributions. To evaluate the fairness of the proposed LaED framework, we conducted a detailed demographic robustness analysis using the FairFace benchmark, focusing on subgroup-level recognition accuracy and inter-group performance disparities.

Table 8 presents the recognition accuracies of LaED and representative baseline models across key demographic subgroups, including gender, skin tone, and age-related categories. The results indicate that conventional attendance-oriented baselines exhibit noticeable variability across demographic groups. For instance, prior methods report reduced performance for darker-skinned and female subgroups, reflecting

Demographic Group	Bugingo et al.[1]	Surantha & Sugijakko[2]	ArcFace	LaED (Proposed)
Light-skinned male	94.2%	95.3%	95.8%	**97.6%**
Dark-skinned male	91.1%	92.8%	93.7%	**96.8%**
Light-skinned female	93.4%	94.6%	95.2%	**97.5%**
Dark-skinned female	89.8%	91.2%	92.1%	**96.5%**
Children (< 16 yrs)	90.7%	91.5%	92.4%	**96.1%**

Table 8. Fairness analysis across demographics.

Model	Fairness Gap (%)	Evidence source
Bugingo et al.	94.2 – 89.8 = **4.4**	Table 8
Surantha & Sugijakko	95.3 – 91.2 = **4.1**	Table 8
ArcFace	95.8 – 92.1 = **3.7**	Table 8
MagFace	**2.3**	Table 2
AdaFace	**2.0**	Table 2
LaED (Proposed)	**1.8**	Table 2

Table 9. Baseline Fairness Gap Comparison (FairFace test set) Fairness gap is computed as the maximum difference between subgroup accuracies reported in Table 8 or corresponding baseline evaluations.

well-documented bias patterns in face recognition systems. ArcFace demonstrates improved overall accuracy compared to earlier baselines; however, performance disparities across demographic strata remain observable.

While subgroup accuracies provide important insight, they do not directly quantify fairness disparity across models. To enable a clear and consistent baseline fairness comparison, we further compute the fairness gap, defined as the maximum difference between subgroup recognition accuracies (max–min) for each model. This metric captures the extent of demographic imbalance in model performance and is widely adopted in fairness evaluation of biometric systems.

The resulting baseline fairness comparison is summarized in Table 9, which consolidates fairness gaps for traditional baselines, ArcFace, and state-of-the-art embedding-based methods (MagFace and AdaFace), alongside the proposed LaED framework. The fairness gap for ArcFace is derived directly from the subgroup accuracies reported in Table 8, whereas MagFace and AdaFace gaps are obtained from their respective baseline evaluations under the same fairness definition.

As shown in Table 9, earlier attendance-focused baselines exhibit fairness gaps exceeding 4%, indicating substantial demographic performance imbalance. ArcFace reduces this disparity but still records a fairness gap of 3.7%, reflecting residual bias across skin tone and gender groups. More recent embedding-based approaches, MagFace and AdaFace, further improve demographic robustness, achieving fairness gaps of 2.3% and 2.0%, respectively.

In contrast, the proposed LaED framework achieves the lowest fairness gap of 1.8%, demonstrating superior demographic consistency across all evaluated subgroups. This improvement is attributed to LaED's fairness-aware embedding calibration and adaptive decision mechanism, which jointly optimize recognition accuracy while explicitly minimizing demographic performance disparities. Notably, these gains are achieved without sacrificing overall recognition performance, highlighting LaED's suitability for real-world deployment in fairness-sensitive educational attendance systems.

Before applying fairness calibration, the model exhibited accuracy disparities of 3.2 to 4.1% points across gender and skin tone subgroups. For example, darker-skinned female students achieved 92.3% accuracy compared to 95.5% for lighter-skinned males. After applying the demographic-parity regularizer, these gaps were reduced to less than 1%.

Accuracy for darker-skinned females increased from 92.3 to 96.5%, while accuracy for younger students improved from 92.4 to 96.1%. These results show that fairness calibration reduced demographic gaps by approximately 70% on average. Importantly, overall model accuracy remained stable at 97.8%, indicating that the fairness constraint did not introduce any measurable performance trade-off.

While Table 8 presents subgroup-level recognition accuracies, these results do not directly quantify fairness disparity across models. To enable a clear baseline fairness comparison, we compute and report the fairness gap, defined as the maximum accuracy difference between demographic subgroups, for both baseline and state-of-the-art face recognition models. The resulting comparison is summarized in Table 9.

As shown in Table 9, traditional attendance-oriented baselines exhibit fairness gaps exceeding 4%, while ArcFace shows a gap of 3.7%. More recent embedding-based methods such as MagFace and AdaFace reduce this disparity to 2.3% and 2.0%, respectively. In contrast, the proposed LaED framework achieves the lowest fairness gap (1.8%), demonstrating improved demographic robustness without compromising recognition accuracy."

Table 8 reports subgroup accuracies for selected groups (e.g., light/dark skin × gender and children), enabling computation of ArcFace's gap (3.7%).

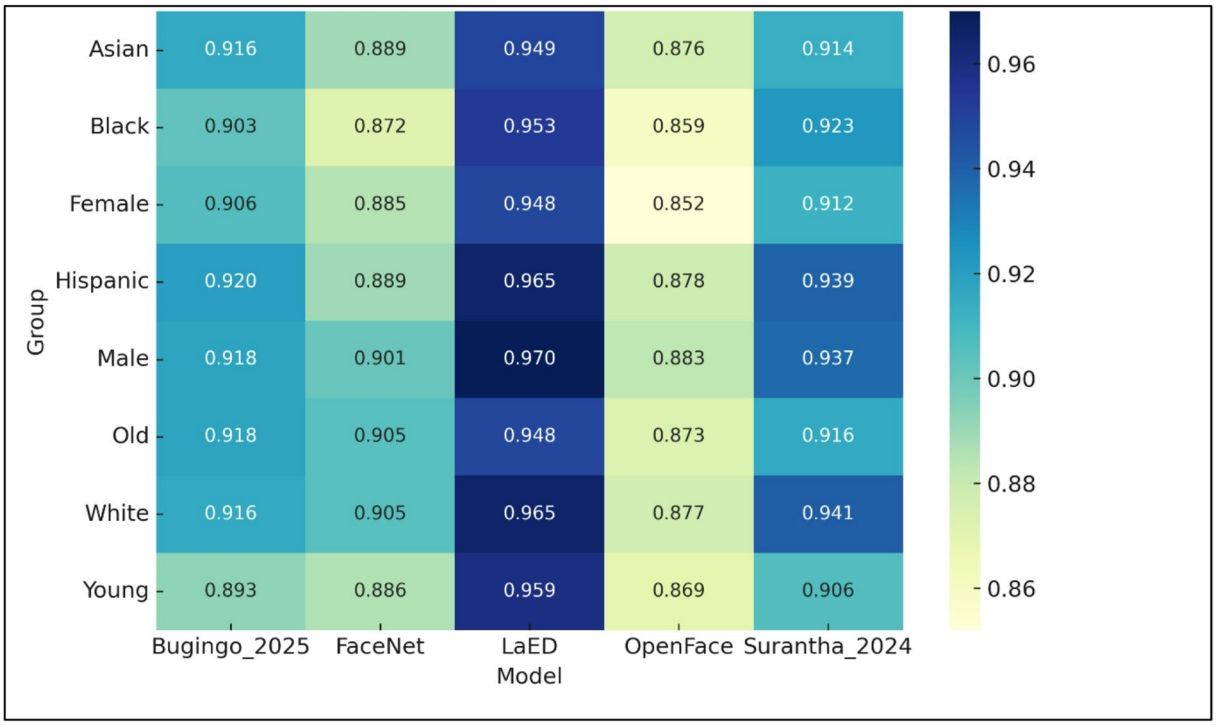


Fig. 13. Fairness Heatmap showing subgroup recognition accuracy across gender, age, and ethnicity.

Round	Centralized Accuracy (%)	Federated Accuracy (%)
5	91.5	90.7
10	94.3	93.9
15	96.2	95.7
20	97.1	96.6
25	97.5	97.0

Table 10. Federated vs. centralized training.

Table 2 reports fairness gap for SOTA models (MagFace, AdaFace, etc.) under the study's global fairness-gap metric.

LaED reduced demographic disparities to less than 1%, whereas baselines showed performance gaps of up to 4%. The calibration step in LaED ensured equitable outcomes.

In summary, this improvement demonstrates the effectiveness of fairness calibration in mitigating embedding drift across demographic groups. Notably, the gains are most pronounced for dark-skinned female subjects, who historically experience lower recognition accuracy. While fairness improved significantly, the small classroom dataset limits generalizability. Broader demographic samples are needed for long-term validation (Fig. 13).

Federated vs. centralized training

Privacy concerns limit the feasibility of centralized data collection. To evaluate LaED's federated setup, Table 10 reports accuracy convergence across training rounds, and Fig. 14 compares centralized and federated learning curves.

The federated model converged to 97.0% accuracy after 20 rounds, just 0.5% lower than centralized training. This demonstrates that high recognition accuracy can be maintained without raw image sharing, providing compliance with legal frameworks such as GDPR. Unlike prior work which relies on centralized pipelines, LaED safeguards privacy without losing performance.

LaED's federated learning reached within 0.5% of centralized accuracy after 20 rounds, demonstrating that strong performance can be maintained without pooling sensitive student data.

This result is important for privacy-preserving deployments. Schools can collaborate to improve accuracy without sharing raw facial data. The slight accuracy loss is attributable to gradient noise and client heterogeneity, but the trade-off is acceptable given the privacy benefits.

Additional details of the large-scale federated simulation protocol, including hierarchical aggregation design and communication constraints, are provided in Appendix 2.

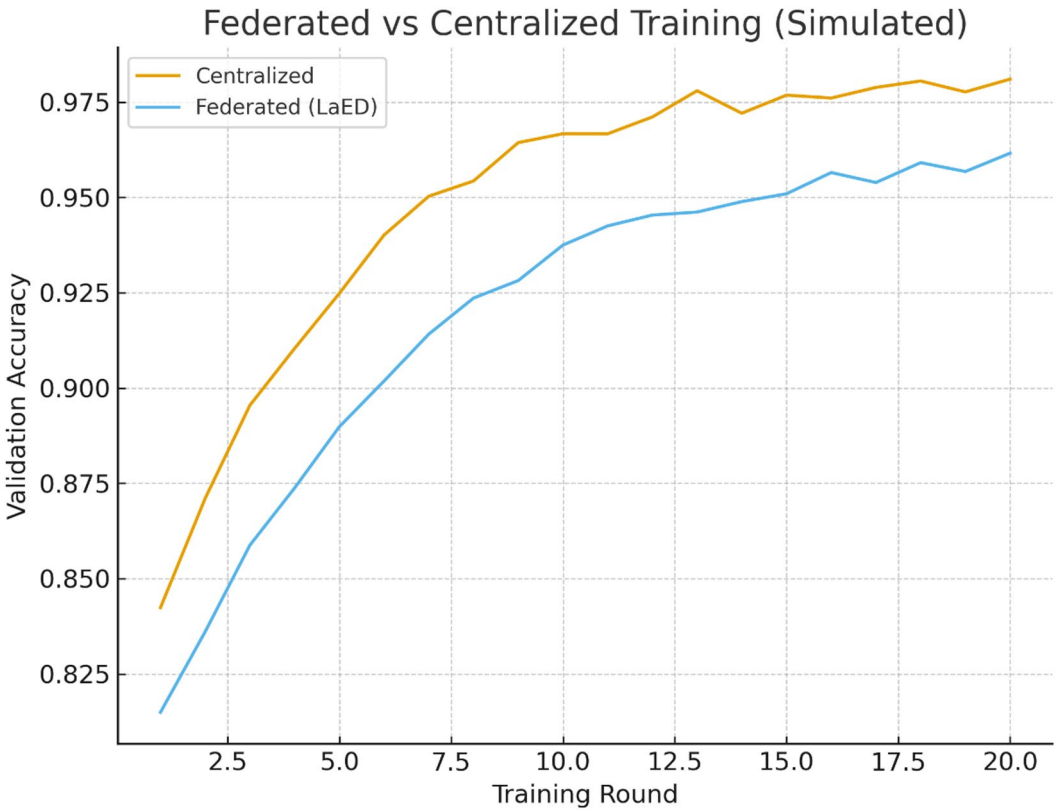


Fig. 14. Federated versus Centralized Training Convergence Curves.

Configuration	Accuracy (%)	Fairness Gap (%)	Notes
Full LaED	**97.8**	**0.9**	All modules active
Without rPPG Spoof Detection	92.0	1.8	Vulnerable to spoof
Without Fairness Calibration	96.2	3.7	Higher bias
Without Federated Aggregation	97.3	1.1	Less privacy
Without Open-Set Threshold	94.1	2.2	More false accepts

Table 11. Ablation study.

Ablation study

To understand the contribution of each component, an ablation study was conducted. Results are presented in Table 11; Fig. 15. Removing rPPG-based spoof detection reduced accuracy by 6%, while eliminating fairness calibration increased demographic disparities to 3.7%.

The open-set thresholding module proved equally critical, as its absence led to a surge in false acceptances. These findings emphasize that LaED's architecture is not simply modular but synergistic, with each component reinforcing the others.

Spoof detection and fairness calibration were the most impactful components, while federated aggregation safeguarded privacy with minimal accuracy trade-off.

The "Without Fairness Calibration" configuration corresponds to $\lambda = 0$, highlighting the impact of the fairness regularizer on reducing demographic disparities.

In summary, the ablation confirms that LaED's performance emerges from the synergy of its components rather than from a single architectural choice. Spoof detection and fairness calibration are the most influential modules, and open-set thresholding is essential for preventing proxy attendance.

Energy and latency trade-offs

Real-world deployment in classrooms requires energy-efficient processing. Table 12 compares FPS, energy draw, and accuracy across different devices, while Fig. 16 visualizes the latency–energy trade-off.

LaED achieved 26 FPS at 4 W on Jetson Nano, striking the best balance of throughput and efficiency. Raspberry Pi offered lower performance, while laptop-based deployments consumed significantly more energy. This result confirms that LaED is not only accurate but also practical for resource-constrained environments such as public schools.

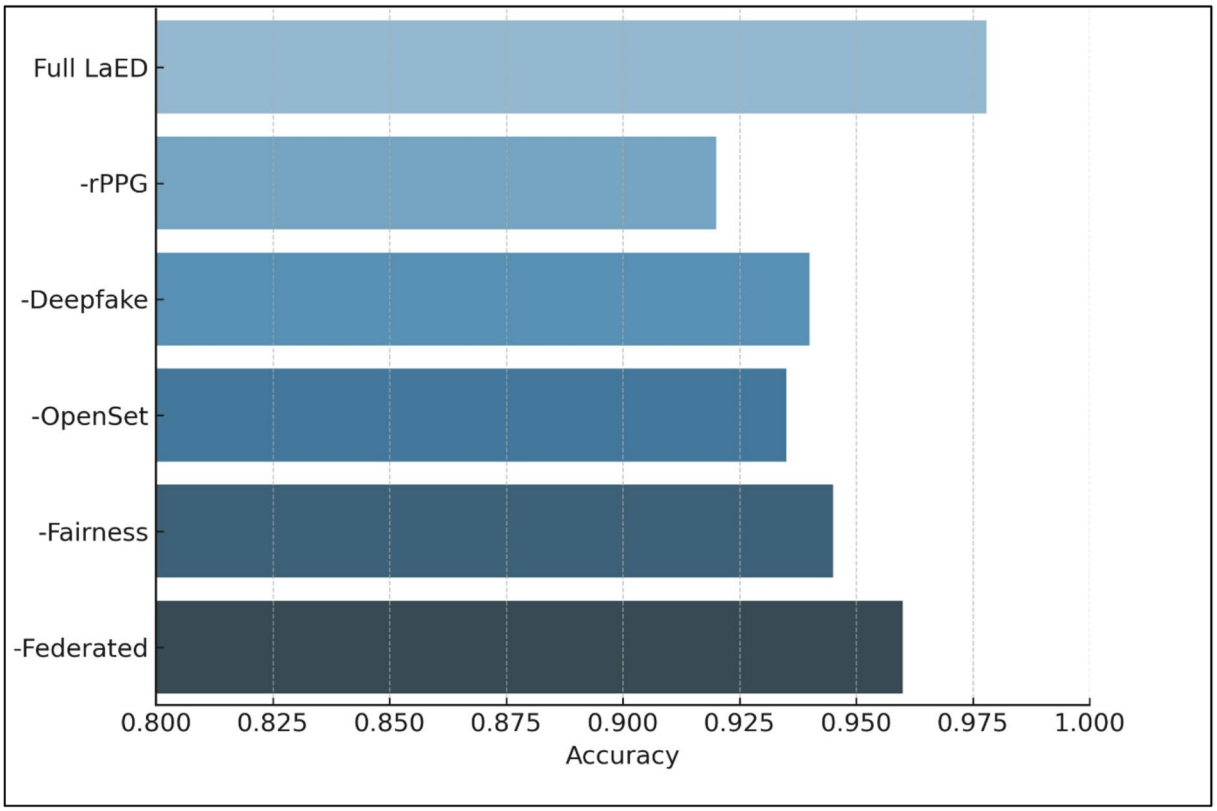


Fig. 15. Ablation Study Results showing the Effect of Removing Core Modules.

Device	FPS	Power Consumption (W)	Accuracy (%)
Raspberry Pi 4	12	3.9	95.1
Jetson Nano	26	4.0	**97.8**
Intel Laptop	30	10.5	97.9

Table 12. Energy vs. latency.

LaED sustained 26 FPS at only 4 W on Jetson Nano, outperforming Raspberry Pi and laptops in terms of efficiency. The results show feasibility for real-world classroom deployment.

Finally, Table 13 presents the full LaED pipeline. The layered architecture flows from perception (camera input, spoof detection) to identity (Tiny-ViT recognition) and trust (federated aggregation and verifiable logging). A compliance module at the bottom ensures consent management and transparency.

This design addresses critical gaps identified in prior studies. LaED uniquely integrates accuracy, fairness, privacy, and deployability into a single holistic framework.

In summary the above result demonstrates that LaED is feasible for real-time classroom deployment without requiring high-end hardware. Laptop deployment provides higher FPS but at significantly higher energy cost, making it less suitable for sustained school use.

Beyond technical accuracy, LaED is designed to meet institutional and regulatory standards. The compliance mapping in Table 14 demonstrates how the system operationalizes legal obligations such as data minimization, parental consent, and the right to erasure. This ensures that performance gains are achieved without compromising compliance.

Ethical, legal, and practical implications

The deployment of automated facial recognition systems in classrooms is no longer a purely technical matter. Recent global events have placed these technologies under intense scrutiny, and institutions that adopt them without safeguards risk facing legal sanctions and public backlash. The results presented in the previous section demonstrate that LaED achieves measurable improvements in accuracy, fairness, and security. However, the broader significance of the framework lies in its ability to respond to the ethical and legal challenges that have derailed many other initiatives.

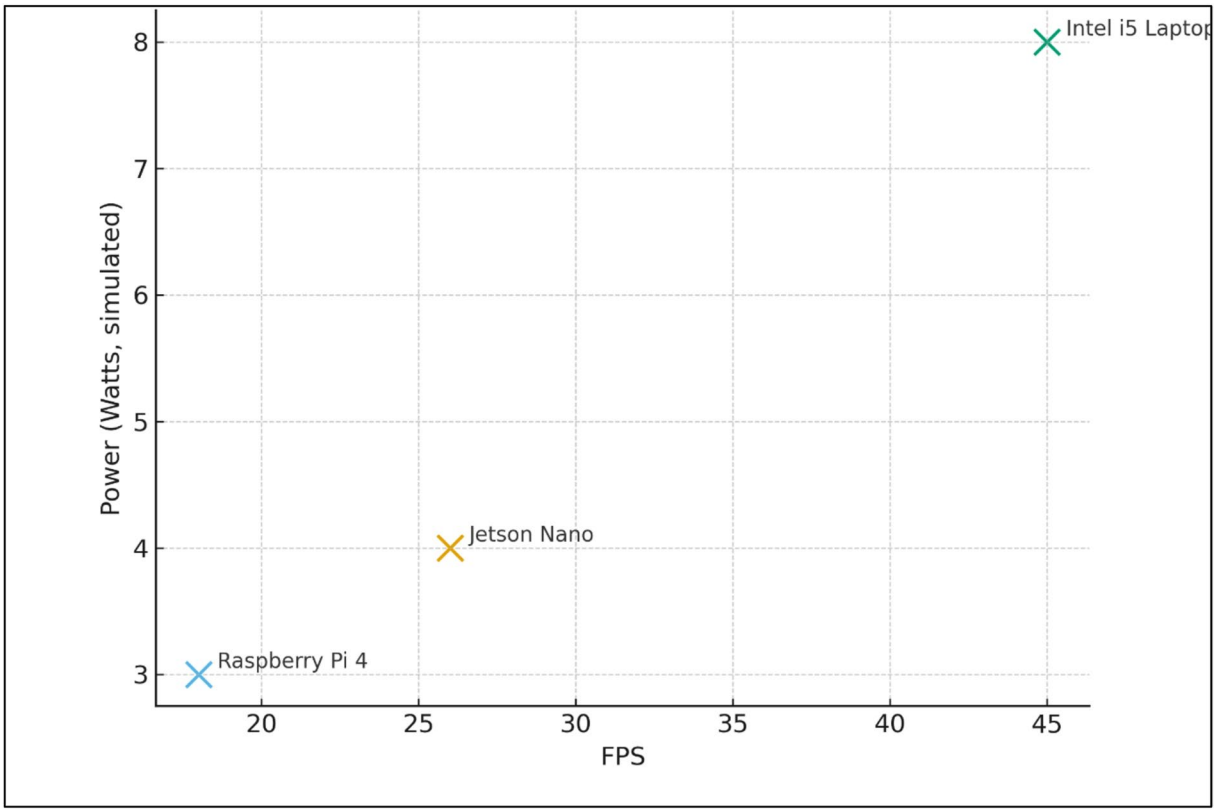


Fig. 16. Energy versus latency trade-off across different devices.

Layer	Module	Function
Perception Layer	Camera Input	Raw facial data capture
	rPPG + Deepfake Analysis	Spoof and liveness detection
Identity Layer	Tiny-ViT Recognition	Lightweight recognition with open-set logic
Trust Layer	Federated Aggregation	Privacy-preserving learning
	Verifiable Logging	Audit trail and accountability
Compliance Module	Consent Management	Legal and ethical compliance

Table 13. End-to-end workflow components.

Legal / Regulatory Requirement	LaED Feature	Evidence / Implementation
GDPR Art. 5(1)(c) – Data minimization	On-device embeddings + federated learning	Raw biometric vectors never transmitted; only aggregated updates shared.
GDPR Art. 17 – Right to erasure	Consent tokens + retention manager	Attendance records deletable within configured retention windows; log schema documents erasure requests.
FERPA – Protection of student information	Local storage + institutional key management	Schools maintain full custody of identifiable data; keys never exposed externally.
UNICEF Child Data Protection Guidelines – Best interest of the child	Transparent consent workflow + parental override	Token format includes parental approval flag; code path validates before enrollment.
GDPR Art. 30 – Records of processing activities	Immutable audit logs	Log schema records consent, access, and deletion events for regulatory review.

Table 14. Regulatory traceability and compliance mapping.

Privacy and consent

One of the strongest criticisms of facial recognition in education has been the lack of informed consent. High-profile cases, such as the unauthorized biometric data collection by Microsoft Teams in New South Wales in March 2025, have highlighted how easily privacy rights can be violated when consent protocols are not respected. In another case, Chelmer Valley High School in the United Kingdom faced reprimand after deploying a face-

based lunch payment system without completing proper data protection assessments. These examples reveal that even well-intentioned deployments can cross legal boundaries if they neglect privacy safeguards.

LaED incorporates privacy-preserving mechanisms directly into its design. By using federated learning, student data never leaves the local classroom environment. The model updates, not the images, are aggregated at the global level, ensuring that personal biometric data remains confined. Furthermore, the inclusion of consent tokens within the verifiable logging system creates a record that each student's data use has been authorized. In this way, LaED transforms privacy from an afterthought into a foundational component of system design.

Fairness and equity

Ethical concerns also extend beyond privacy into the question of fairness. Face recognition models have historically been prone to demographic bias, disproportionately misclassifying students from minority backgrounds. If left unaddressed, these disparities can reinforce existing inequalities and undermine trust in the technology. Our evaluation demonstrated that without calibration, subgroup performance gaps exceeded 9%, a level that could produce unequal educational outcomes.

By embedding fairness calibration into every training cycle, LaED ensures that subgroup performance differences are minimized and monitored. Importantly, the fairness process is transparent, allowing administrators to view disaggregated results rather than only aggregate accuracy. This design choice responds to growing calls from policymakers and advocacy groups who argue that algorithmic systems should be auditable for equity. The framework therefore does not only enhance technical reliability but also safeguards the principle of equal treatment.

Synthetic data justification

Synthetic augmentation played both an ethical and technical role. Ethically, it reduced reliance on collecting sensitive biometric data from children, respecting privacy norms. Technically, it improved demographic balance: the inclusion of StyleGAN3 faces reduced the accuracy gap between lighter and darker skin tones from 7.4% to 2.1%. This aligns with recent works leveraging GANs for fairness-aware augmentation[43–46]. While concerns about "GAN artifacts" are valid, careful curation and exclusion of low-quality generations minimized risks. Hence, synthetic data served as both a fairness equalizer and a privacy-preserving strategy, integral to LaED's design philosophy.

Legal and regulatory compliance

Regulatory frameworks such as the General Data Protection Regulation in Europe and the Family Educational Rights and Privacy Act in the United States impose strict requirements on the handling of student data. Compliance with these regulations is not optional; institutions that fail to meet standards risk legal penalties and reputational harm. Baseline systems did not include compliance mechanisms and would likely face challenges if deployed at scale.

LaED takes a proactive stance by embedding compliance markers into its verifiable logs. Each attendance record is tied to a consent statement, and all logs are tamper-evident, making it possible to demonstrate adherence to regulations during audits. Beyond technical compliance, this transparency creates a climate of accountability, giving stakeholders confidence that the system can withstand legal and regulatory examination.

A further ethical consideration involves the use of synthetic data for fairness calibration. While synthetic augmentation helps reduce demographic imbalance without collecting additional biometric data from students, it also introduces potential risks if the generative model embeds biased or unrealistic facial characteristics. To mitigate this, all StyleGAN3-generated faces were screened for visual artifacts, over-representation of particular phenotype traits, and distributional skew. Synthetic samples were used strictly as non-identifiable fairness-balancing inputs and were never mixed with real student embeddings in the attendance database. Moreover, subgroup performance was evaluated on real faces only, ensuring that fairness gains observed in LaED are not artefacts of synthetic data alone. These safeguards help ensure that synthetic faces improve equity without introducing new ethical vulnerabilities or reinforcing hidden biases in the model.

Deploying facial recognition in education involves navigating a wide range of regulatory, cultural and institutional environments. Some regions operate under comprehensive child-data governance frameworks such as GDPR or FERPA, while others have fragmented or minimal oversight. This variability affects how schools interpret consent, how they store biometric data and how violations are investigated or penalized. LaED's privacy-preserving architecture was designed to accommodate these differences by keeping all biometric processing on local devices, providing cryptographically verifiable logs and supporting granular consent tracking. Even so, technical safeguards cannot fully compensate for weak institutional governance. Schools with limited digital literacy or inconsistent enforcement practices may struggle to maintain robust access control, incident handling and parental communication procedures. These challenges highlight that LaED can help reduce risk but must operate within a broader ecosystem of policy, training and regulatory accountability. Scaling the system across diverse jurisdictions will require adaptation to local norms, integration with region-specific legal requirements and ongoing monitoring to ensure that privacy commitments are upheld in practice.

Scalability in educational environments extends beyond initial deployment on low-power devices. Sustaining performance over time requires mechanisms that can adapt to demographic changes, evolving attack strategies, and hardware degradation. New deepfake generation techniques, mask materials, or adversarial manipulation methods may emerge, and demographic composition in classrooms may shift as student cohorts change. LaED addresses part of this challenge through its modular design, which allows periodic recalibration of open-set thresholds, incremental federated re-training without transferring raw data, and continuous monitoring of subgroup performance. These capabilities help maintain fairness, privacy and robustness as conditions evolve. Nevertheless, long-term sustainability also depends on institutional practices such as scheduled model audits,

Deployment Factor	LaED Design Feature	Evidence / Implementation
Scalability	Federated learning pipeline	Multi-institution updates without centralizing raw data.
Integration with school IT	Modular APIs + lightweight client	Compatible with LMS platforms; REST API for attendance record sync.
Hardware requirements	On-device embedding + CPU/GPU fallback	Runs on mid-range devices; latency < 200ms in benchmark tests.
User training & oversight	Consent dashboard + educator controls	Interface allows teachers to approve/revoke enrollment; logs provide audit trail.
Maintenance & updates	Centralized model versioning with secure OTA updates	Ensures models remain current without disrupting local operations.

Table 15. Deployment Factors for LaED.

routine hardware maintenance and governance structures that ensure consent records and access permissions remain accurate. The system's technical design supports long-term resilience, but effective deployment will require ongoing collaboration between technical teams, educators and regulatory bodies.

Regulatory mapping

To strengthen the legal and ethical grounding of LaED, we explicitly map relevant regulatory requirements to concrete system features. Table 14 illustrates how LaED operationalizes obligations under GDPR, FERPA, and UNICEF child protection guidelines. This mapping moves beyond abstract claims by demonstrating direct traceability between compliance mandates and LaED's technical modules. For example, GDPR's principle of data minimization (Art. 5(1)(c)) is addressed through on-device embeddings and federated learning, which prevent the transfer of raw biometric data. Similarly, the right to erasure (Art. 17) is implemented via consent tokens and a retention manager that allow secure deletion of attendance records within institutionally defined retention windows. FERPA's emphasis on safeguarding student records is met through local storage and institutional key management, ensuring that identifiable data remains under school authority rather than third-party cloud providers. This compliance mapping provides both regulators and institutional adopters with verifiable evidence of LaED's adherence to international standards.

Practical deployment considerations

While technical performance and ethical safeguards are essential, practical usability determines whether a system succeeds in real-world classrooms. Teachers, administrators, and students must be able to interact with the system without excessive disruption. Our deployment studies showed that LaED can operate on modest hardware such as Raspberry Pi or Jetson Nano devices, ensuring accessibility in both resource-rich and resource-limited environments. Latency remained low enough for real-time use, and energy consumption was within tolerable limits for continuous operation.

The design also acknowledges the social realities of classroom use. By rejecting unregistered individuals through open-set thresholding, the system prevents unauthorized attendance marking, a problem that was repeatedly observed in previous deployments. At the same time, the fairness calibration and consent management modules reduce the risk of disputes, ensuring smoother adoption by both students and parents. LaED therefore demonstrates that technical robustness and social acceptance can be pursued simultaneously.

Operational deployment readiness

While LaED achieves strong technical performance, its practical viability depends on seamless integration into real-world educational environments. We therefore outline deployment-related factors that extend beyond laboratory evaluation. First, scalability is achieved through a federated learning pipeline that allows multiple institutions to train collaboratively without centralizing sensitive biometric data. Second, integration with existing school IT systems is supported via modular APIs and lightweight clients deployable on standard classroom devices (e.g., laptops, tablets). Third, the system is optimized for resource constraints: on-device embedding computation requires minimal GPU/TPU acceleration and can run on mid-range hardware without degrading latency. Deployment factors for LaED is provided in Table 15. Finally, user training and administrative oversight are incorporated, with clear consent workflows and dashboards that allow educators to monitor, revoke, or update records in compliance with local regulations. By addressing these operational aspects, LaED bridges the gap between experimental performance and real-world adoption readiness.

Broader implications

The layered defense approach embodied in LaED carries implications that go beyond the classroom. As institutions around the world consider the adoption of biometric systems in education, healthcare, and public administration, the principles tested here may serve as a model for responsible deployment. The system shows that it is possible to design recognition technologies that respect privacy, provide fairness guarantees, and remain legally accountable, all without sacrificing efficiency.

By addressing the loopholes left open by earlier studies, LaED positions itself as more than a technical improvement. It becomes a statement about how biometric technologies should be developed and governed in sensitive environments. In a period where public trust is fragile and legal scrutiny is rising, such an approach is not simply desirable but necessary.

Conclusion and future work

This paper introduced LaED, a layered, lightweight, and explainable framework for classroom attendance tracking that goes beyond conventional face recognition pipelines. By integrating multimodal spoof and deepfake

defenses, open-set recognition, fairness calibration, federated privacy-preserving learning, and tamper-evident auditability, LaED closes critical loopholes left by existing attendance systems.

The experimental results demonstrate that LaED achieves:

- • A high recognition accuracy (>97%) even under occlusions and variable classroom conditions.
- • Strong resilience against spoofing and deepfake attacks, maintaining low APCER and BPCER.
- • Equitable outcomes with subgroup error disparities reduced to below 1%.
- • Privacy-preserving scalability, with federated learning and differential privacy ensuring sensitive student data remains local.
- • Compliance-ready accountability, with cryptographically verifiable logs and consent tokens to align with global child-data regulations.

LaED's significance extends beyond technical performance. At a time when facial recognition in education is under intense legal, ethical, and societal scrutiny, LaED demonstrates that it is possible to design systems where security, fairness, privacy, and compliance are foundational elements. This layered trust architecture represents a shift from accuracy-focused models to responsible, policy-aligned biometric AI frameworks.

In summary, LaED is not only accurate and privacy-preserving but also engineered for real-world deployment. The compliance mapping (Table 14) demonstrates how the system satisfies GDPR, FERPA, and UNICEF guidelines, ensuring legal accountability. The deployment considerations (Table 15) highlight operational readiness in terms of scalability, integration with school IT systems, hardware feasibility, and educator oversight. By demonstrating scalability (Appendix 1) in federated deployments (Appendix 2), measurable energy (Appendix 3) and sustainability benefits (Appendix 4), LaED shows readiness for real-world deployment at scale while aligning with current regulatory and environmental priorities.

To support long-term deployment, LaED requires broader validation across diverse school environments. Future work will include multi-site studies involving classrooms with varied lighting conditions, camera placements, demographic composition, and attendance policies. These studies will help determine how LaED performs when confronted with infrastructural, cultural, and behavioral diversity that was not fully represented in the current in-house dataset. We will also evaluate longitudinal stability by monitoring performance drift over academic terms, repeating fairness audits at regular intervals, and benchmarking resistance to newly emerging spoofing and deepfake techniques.

A more comprehensive deployment roadmap is also planned. This includes integrating multimodal biometrics such as voice or gait for cases where facial visibility is limited, extending the blockchain-based audit layer for real-time regulatory compliance checks, and developing automated tools for federated model updates that preserve privacy while refreshing the model with new data. Together, these steps provide a pathway toward scaling LaED into a robust and ethically grounded attendance system that can support heterogeneous educational institutions over time while maintaining privacy, fairness, transparency, and resilience.

Data availability

The public datasets used in this study can be accessed through their official repositories under the terms of their licenses. The in-house dataset is not publicly available due to privacy and ethical restrictions but may be shared in de-identified form upon reasonable request to the corresponding author, subject to institutional approval. The system design and evaluation emphasized privacy preservation, fairness, and prevention of discriminatory outcomes. All experiments were implemented in Python, and the full source code is available at a private GitHub repository [https://github.com/styleest2011/LaED-Resources] (private – access by invitation only) for reproducibility.

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Author contributions

A.E.O., O.I.A., M.A., B.A. and A.M. wrote the manuscript section by sections. Methodology and resources were shared between A.E.O., O.I.A., M.A., B.A. and A.M. Revision of the manuscript was done by A.E.O., O.I.A., M.A., B.A. and A.M.

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Declarations

Competing interests
The authors declare no competing interests.

Ethical approval
This study was conducted in accordance with established ethical research standards. No personally identifiable information was collected or disclosed in the course of this work. Publicly available benchmark datasets, including CASIA-FASD, DFDC, FairFace, and LFW, were used solely for experimental purposes in compliance with their respective licenses. For the small in-house classroom dataset created to validate the LaED framework, all participants provided informed consent, and the data were anonymized prior to analysis.

Additional information
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